

THE EUROPEAN CREWMAN'S PRIMER

Conduct, Readiness, and Survival in the Submariner's Trade

Issued for circulation aboard commercial, Coalition, research, transport, salvage, and independently operated vessels beneath the European ice.

FOREWORD

There are easier ways to earn a living than serving aboard a submarine on Europa.

There are warmer ways. Safer ways. Ways that do not place several thousand tonnes of steel between crushing pressure on one side and a temperamental reactor on the other. Ways that do not require a person to share narrow corridors with armed security officers, exhausted engineers, overworked medics, unstable assistants, and whatever has begun scratching at the outer hull.

Nevertheless, you have chosen the submariner's trade.

Perhaps you came for the marks. Perhaps you are running from debts, authorities, relatives, former employers, or all four at once. Perhaps you genuinely believe there is something noble in carrying supplies between the last lights of human civilization. Whatever brought you below the ice, your survival now depends upon the people beside you—and theirs depends upon you.

A submarine is not merely a vehicle. It is a sealed and interdependent community operating inside an environment that is instantly lethal to human life. Every hatch, junction box, oxygen tank, ammunition crate, medical cabinet, and pressure seal exists because someone may need it during the worst minute of their life.

Competence matters. Discipline matters. Restraint matters.

Most importantly, cooperation matters.

This primer does not demand perfection. Every veteran was once an inexperienced hand, and every capable officer has made mistakes they would prefer omitted from the official record. What is required is that each crew member make a reasonable effort to learn, communicate, conserve resources, follow lawful orders, and avoid placing the vessel in unnecessary danger.

Europa is already trying to kill us.

The crew need not assist it.

REGARD EVERY CONTROL AS LOADED

A newly assigned crew member will encounter switches, levers, consoles, detonators, discharge controls, reactor panels, weapon interfaces, ballast controls, docking systems, and emergency devices whose purpose may not be immediately obvious.

Do not operate unfamiliar equipment merely to discover what it does.

A switch aboard a European submarine may control a lamp. It may also discharge an electrical coil through a diver working outside the hull. A control at the helm may alter ballast, disengage autopilot, open a docking hatch, change the vessel's depth, or send the submarine directly into rock.

The consequences of careless experimentation include, but are not limited to:

- Reactor instability or meltdown.
- Accidental discharge of nuclear or conventional ordnance.
- Electrocuting of personnel outside the vessel.
- Uncontrolled flooding or ballast failure.
- Collision with terrain, wreckage, ice, or hostile fauna.
- Destruction of a shuttle attempting to dock.
- Loss of weapons required later in the voyage.
- Sudden and entirely preventable death.

Before operating unfamiliar equipment, ask a qualified crew member what it controls. Observe how the system is used. Consult available labels, diagrams, manuals, or status displays.

Curiosity is valuable. Uncontrolled curiosity aboard a pressure vessel is not.

When in doubt, leave the control untouched until you understand it.

REPORT ABOARD READY TO LEARN

No captain expects a new submariner to know every vessel, every junction route, or every emergency procedure. Submarines differ. Captains differ. Some engineers have rewired their vessels so thoroughly that not even the original shipyard could explain them.

What the crew may reasonably expect is that you understand the fundamental controls of your own body and profession.

Before joining an active voyage, learn how to:

- Move through compartments and climb ladders.
- Open, close, and secure hatches.
- Pick up, equip, store, and transfer items.
- Use the vessel's inventory and item-finding systems.

- Wear a diving suit and exchange oxygen tanks.
- Operate the basic tools assigned to your profession.
- Communicate over radio and local voice channels.
- Recognize flooding, fire, low oxygen, and reactor alarms.
- Identify the principal workspaces aboard a submarine.

The middle of a hull breach is not the appropriate time to ask how to equip a welding tool. A reactor fire is not the ideal setting for a detailed explanation of electrical repair. A medic treating three casualties cannot be expected to abandon them because someone has misplaced the screwdriver already attached to their tool belt.

Ask questions, but exercise judgment regarding when and how you ask them.

Questions concerning an unfamiliar submarine, an unusual system, a mission objective, or a genuine shortage are entirely reasonable. Experienced crew members should answer them when circumstances permit. No one benefits when a new hand remains silent, guesses incorrectly, and creates a larger emergency.

Before reporting an item missing, use the item finder and search the appropriate storage compartments. Before requesting personal instruction during a crisis, determine whether the matter can wait until the vessel is stable.

Self-sufficiency does not mean knowing everything. It means attempting the obvious solutions before transferring the entire burden to someone else.

New personnel should be taught.

New personnel must also make an honest effort to learn.

THE GENERAL CREWMAN

The Assistant, the Apprentice, and the Person Somehow Doing Everyone Else's Job

If you are joining a submarine for the first time and cannot yet meet the basic requirements of a specialist position, you should strongly consider signing aboard as an assistant.

This is not an insult.

It is not an admission that you know nothing.

It is not an announcement that you are useless, expendable, or expected to spend the voyage sitting inside a storage cabinet until something needs testing.

It means you are entering the crew as a general submariner rather than falsely claiming expertise in medicine, engineering, security, or navigation.

That is sensible.

It is certainly more sensible than signing aboard as a medical officer because you once successfully applied an adhesive bandage, or becoming a reactor engineer because you have seen electricity and found the experience memorable.

The assistant is the submarine's flexible crewman: a worker who can support every department, learn several trades, fill sudden gaps, and take responsibility for the dozens of important tasks that specialists are too busy, injured, flooded, or on fire to perform.

A new assistant is effectively saying:

"I do not yet know everything, but I am willing to work, learn, carry things, follow instructions, and refrain from lying about my qualifications."

On Europa, this places you ahead of an alarming percentage of the labor market.

YOU ARE NOT THE LEFTOVER ROLE

Specialist positions aboard a submarine are limited.

There may already be a captain, doctor, engineer, mechanic, and security officer. This does not mean the important jobs have been claimed and you have arrived too late to contribute.

It means the obvious jobs have been claimed.

The less obvious jobs remain everywhere.

Somebody still needs to:

- Carry ammunition to the guns.
- Bring welding fuel to the mechanic.
- Bring electrical components to the engineer.
- Drag wounded personnel to medical.
- Refill oxygen tanks.
- Retrieve tools.
- Move supplies.
- Watch the rear of the away team.
- Operate an unattended pump.
- Extinguish a fire.
- Report the creature entering the ballast tank.
- Explain why there is a creature entering the ballast tank.
- Help remove the creature from the ballast tank.
- Prevent someone from putting another creature in the ballast tank to see whether they become friends.

The assistant is not what remains after every useful role has been selected.

The assistant is the role that remains useful after every specialist becomes occupied.

A submarine with no assistants may still operate.

It will simply do so more slowly, less efficiently, and with every specialist periodically abandoning critical work to search for a screwdriver.

THE ASSISTANT'S TRUE SPECIALTY

An assistant is often described as lacking a specialty.

This is inaccurate.

The assistant's specialty is whatever has become a problem in the last thirty seconds.

A specialist usually owns a clearly defined responsibility.

The engineer manages the reactor and grid.

The mechanic maintains the hull and machinery.

The doctor treats the injured.

Security shoots creatures while attempting not to shoot the people standing in front of the creatures.

The captain points the submarine in the direction least likely to contain a wall.

The assistant's responsibilities are more flexible:

- If something needs carrying, carry it.
- If something needs fetching, fetch it.
- If someone needs help, help them.
- If a compartment is flooding, find a pump or a welding tool.
- If a gun is empty, bring ammunition.
- If someone is bleeding, move them toward the person wearing the medical uniform.
- If the person wearing the medical uniform is the one bleeding, move them anyway.
- If the captain asks where the away team is, provide a more useful answer than "away."

The assistant goes where the submarine's current emergency has created a labor shortage.

This is why a veteran assistant may be more useful than a mediocre specialist.

The specialist knows one department.

The veteran assistant knows where every department keeps the things it claims nobody else is allowed to touch.

YOU ARE NOT USELESS

New assistants occasionally believe that they are unimportant because no machinery has their name written above it.

Do not make this mistake.

The reactor has an engineer.

Medical has a doctor.

The helm has a captain.

The assistant has the entire remaining submarine.

You are not ballast with pockets.

You are not an emergency ration for the mudraptors.

You are not the crew member automatically selected to touch every suspicious artifact because “the specialists are too valuable.”

You are certainly not required to wear a clown mask, although historical precedent suggests one may eventually appear in your locker whether you requested it or not.

A competent assistant reduces the amount of time specialists spend doing work that does not require specialist training.

Consider the following:

A mechanic is welding a major hull rupture. Their welding tank empties.

Without an assistant, the mechanic stops repairing the hull, runs across the submarine, searches for fuel, returns, and discovers that the sea has used the delay to purchase the compartment.

With an assistant, a fresh tank arrives before the first one empties.

The mechanic keeps welding.

The hull remains intact.

The crew survives.

The assistant did not perform the dramatic repair.

They made the dramatic repair possible.

This distinction describes a great deal of real work.

THE MULTIPLIER OF PEOPLE WHO ALREADY LOOK BUSY

A heavily experienced assistant can multiply the effectiveness of every department aboard the vessel.

Not literally.

If your assistant begins physically multiplying, notify medical and do not allow either copy near the ballast controls.

Operationally, however, a capable assistant enables specialists to remain at their stations.

The engineer stays at the reactor because the assistant brings replacement components.

The mechanic keeps sealing breaches because the assistant refills welding tanks.

The medic continues treating critical casualties because the assistant carries patients into medical.

Security keeps firing because the assistant reloads the weapon.

The captain remains at the helm because the assistant checks whether the away team has returned.

This is the assistant's real value:

They prevent skilled personnel from spending half the emergency running errands.

A veteran general crewman may walk into a damaged compartment, identify what the specialist is about to need, retrieve it, and return before being asked.

This is not telepathy.

It is experience, observation, and familiarity with the recurring habits of professionals who always discover they need another component when the nearest cabinet is three rooms away.

THE SACRED DUTY OF FETCHING THINGS

Fetching supplies is sometimes dismissed as unimportant work.

It becomes considerably more important when the item being fetched is:

- The last welding tank.
- A replacement fuel rod.
- The blood pack required by a dying crew member.
- Ammunition for the only gun facing the threat.
- A diving suit for the person trapped inside a flooding compartment.
- A fire extinguisher.
- The captain's emergency revolver.
- The engineer who knows why the reactor has begun making that noise.

A good assistant learns where supplies are stored.

A great assistant learns where supplies are actually stored after the crew has finished placing them anywhere except their designated storage.

The labels may say:

- Welding equipment.
- Medical supplies.
- Electrical components.
- Ammunition.
- Diving equipment.

The truth may be:

- Welding equipment in medical.
- Medical supplies in the captain's cabin.
- Electrical components in a toolbox carried by an unconscious mechanic.
- Ammunition scattered across three compartments.
- Diving equipment worn by someone who went to the station bar.

Finding things is a legitimate maritime skill.

The assistant who knows where the replacement fuel rods are may briefly become the most important person on the submarine.

Enjoy the moment.

Then bring the rods to engineering.

LEARN THE BOAT BEFORE THE BOAT LEARNS YOU

The assistant position is one of the best ways to learn the layout of a submarine.

Specialists frequently remain near their assigned departments.

The assistant travels everywhere.

Use this opportunity to learn:

- Where the reactor is.
- Where the replacement fuel is.
- Where the engine is.
- Where the pumps are.
- Which rooms are ballast tanks.
- Where oxygen is generated.
- Where medical stores its supplies.

- Where security stores weapons.
- Where ammunition is supposed to be stored.
- Where ammunition has actually been stored.
- Where the airlocks are.
- Which airlock someone has left open.
- Where the fabricator and deconstructor are.
- Which doors lead toward safety.
- Which doors lead into a flooded ballast tank.
- Which controls should never be operated merely because they are red and pleasingly illuminated.

An assistant who learns the submarine can respond quickly.

An assistant who does not may spend the emergency running through the same corridor repeatedly while asking where engineering is.

The third time you pass the reactor, consider the possibility that you have found it.

THE ASSISTANT'S UNIVERSAL RESPONSE KIT

A general crewman should gradually learn how to use the basic tools found throughout the vessel.

A practical assistant may carry or know how to obtain:

- A wrench.
- A screwdriver.
- A welding tool.
- A spare welding tank.
- A fire extinguisher.
- A diving mask.
- A basic medical item.
- A flashlight.
- A radio headset.
- A tool belt.
- Enough empty storage space to carry something useful.

Do not carry every tool aboard the submarine.

The objective is to remain prepared, not to become a mobile evidence locker.

An assistant carrying three welding tools, two crowbars, four masks, every screwdriver aboard, nine bandages, an empty shotgun, two fuel rods, and a musical instrument has not created an effective loadout.

They have become a loot container capable of speech.

Take what you need for the work you are performing.

Return what you no longer require.

Leave enough equipment for the other people attempting to keep the vessel alive.

ASK WHAT NEEDS DOING

When unsure how to contribute, ask the department heads.

Useful questions include:

- “Engineering, do you need components?”
- “Mechanics, where is the worst breach?”
- “Medical, do you need casualties moved?”
- “Security, which gun needs ammunition?”
- “Captain, where do you need another pair of hands?”
- “Away team, do you need a porter or lookout?”
- “Who used the last welding tank?”
- “Why is there water coming through the ceiling?”
- “Was the ceiling always supposed to do that?”

Ask once.

Listen to the answer.

Do the work.

Do not stand beside a mechanic actively welding the ocean out of the compartment and repeatedly ask whether they need help while making no effort to retrieve the empty tank at their feet.

If everyone is too busy to answer, look for obvious work:

- Refill tanks.
- Return tools.
- Recharge batteries.
- Sort supplies.
- Carry ammunition.
- Clear the corridors.
- Extinguish fires.
- Check common compartments for damage.
- Move salvage.
- Prepare diving equipment.
- Watch the route nobody else is watching.
- Stop the assistant beside you from pressing the emergency discharge button “for morale.”

Initiative is valuable.

Unsupervised experimentation is merely initiative with a casualty report attached.

DO NOT HELP SO HARD THAT YOU BECOME THE PROBLEM

There is such a thing as excessive assistance.

If an engineer is repairing a junction box, you do not need to stand inside the engineer.

If a medic is treating one casualty, seven assistants do not need to gather around and offer conflicting medical theories.

If a mechanic is welding a breach, do not block their path to the exit.

If security is firing through a doorway, do not run through the doorway because you have brought them ammunition.

A helpful assistant:

1. Determines what is needed.
2. Brings it.
3. Communicates clearly.
4. Moves out of the way unless further assistance is required.

A less helpful assistant brings the wrong object, places it on the floor, blocks the hatch, asks whether anyone needs a harmonica, and is then surprised when the compartment floods.

Help should reduce chaos.

If your contribution increases the number of people shouting, reassess the contribution.

SHADOWING SPECIALISTS WITHOUT BECOMING THEIR SHADOW

Assistants are encouraged to observe experienced specialists during quiet periods.

This is how general crewmen develop into capable engineers, mechanics, medics, security officers, and captains.

Ask:

- How does the reactor work?
- How do pumps fail?
- Which injuries require immediate treatment?
- How is ammunition loaded?
- How does the captain distinguish a mineral formation from a large creature?

- Why does the electrical engineer look tired even when nothing is happening?

Choose the correct time.

The engineer may be happy to explain fission rates after the reactor stops overheating.

The doctor may demonstrate diagnostic procedure after the patient stops losing blood.

The captain may explain sonar after avoiding the wall currently occupying most of it.

A useful phrase is:

“When things are stable, could you show me how that works?”

An unhelpful phrase is:

“What does this button do?”

Especially when your finger is already pressing it.

BECOMING DANGEROUSLY COMPETENT

A capable assistant should gradually learn several basic skills.

These may include:

- Repairing small hull breaches.
- Operating pumps.
- Replacing oxygen tanks.
- Extinguishing fires.
- Applying bandages.
- Moving unconscious crew members.
- Reloading common weapons.
- Repairing simple mechanical or electrical damage.
- Using diving equipment.
- Recognizing alarms.
- Reporting emergencies properly.
- Fabricating routine supplies.
- Carrying salvage without wandering into another cave.
- Holding the helm steady if every qualified officer has become temporarily horizontal.

The objective is not to replace specialists.

It is to keep the problem from worsening until the specialist arrives.

A basic repair performed promptly may preserve a system.

A bandage may keep someone alive.

A pump switched on at the correct moment may save a compartment.

A door closed during flooding may save half the vessel.

The assistant's greatest victories often appear very small.

They are only small because they prevented the larger disaster.

ASSISTANTS DURING FLOODING

Flooding is where a useful assistant earns the undying affection of mechanical engineering, or at least a temporary reduction in the amount of swearing directed at them.

When water enters the submarine:

- Locate the breach.
- Inform the crew.
- Close doors to slow the spread.
- Put on appropriate equipment.
- Retrieve a welding tool.
- Assist the mechanic.
- Operate or repair the pump.
- Watch for creatures entering through the opening.
- Avoid sealing yourself on the wrong side of the door.
- Avoid sealing the mechanic on the wrong side of the door.
- Avoid declaring the compartment secure while the water reaches your helmet.

If the breach is beyond your ability, report it and support the person repairing it.

Do not continue welding a tiny crack in an empty storage room while the reactor compartment becomes an indoor sea.

Priorities matter.

The ocean will patiently wait for you to choose incorrectly.

ASSISTANTS DURING COMBAT

In combat, assistants may:

- Reload guns.
- Move ammunition.

- Carry medical supplies.
- Transport casualties.
- Seal damaged compartments.
- Repair pumps.
- Watch uncovered approaches.
- Support security.
- Operate a weapon if trained and assigned.
- Refrain from standing in the line of fire.

Do not seize the largest weapon aboard because the sound of combat has inspired you.

Do not fire through crew members.

Do not open the armory and distribute weapons to everyone with pockets.

Do not approach security during an active firefight by silently running toward them while holding an unannounced firearm.

If you must cross a firing line, communicate.

If you are carrying ammunition, say so.

If you see a hostile creature, report its location rather than yelling “THING” over the radio.

Europa contains a very large number of things.

Be specific.

ASSISTANTS IN MEDICAL

A general crewman does not need to become a doctor to help medical personnel.

Useful medical support includes:

- Carrying casualties.
- Applying simple bandages.
- Retrieving blood packs.
- Bringing oxygen.
- Clearing the treatment area.
- Following the doctor's instructions.
- Preventing untreated patients from wandering away.
- Informing medical what caused the injury.
- Refraining from administering every drug in the cabinet because the patient “looked poorly.”

The phrase “I found them like this” is less useful than:

“They were bitten by a mudraptor, lost blood, and were unconscious for approximately thirty seconds.”

Do not perform experimental surgery.

Do not prescribe morphine because you like the patient.

Do not prescribe morphine because you dislike the patient.

Do not prescribe morphine because the ampoule is already in your hand.

The assistant helps the doctor work.

The assistant does not establish a competing medical practice in the corridor.

ASSISTANTS IN ENGINEERING

Engineering departments adore a competent assistant for the same reason exhausted parents adore anyone willing to carry groceries.

Useful engineering support includes:

- Delivering components.
- Refilling welding tanks.
- Moving fuel rods.
- Sorting fabrication materials.
- Repairing secondary junction boxes.
- Watching pumps.
- Relaying reactor status.
- Helping drain compartments.
- Recovering tools.
- Keeping other crew members away from controls they do not understand.

Do not change reactor settings without instruction.

Do not add fuel rods because “more fuel means more power.”

Do not rewire systems because you have discovered the wiring tool.

Do not unplug the oxygen generator during an ordinary voyage to test the silent-running chapter.

The engineer may teach you these systems over time.

Until then, your most valuable electrical skill may be locating the qualified electrical engineer.

ASSISTANTS ON AWAY MISSIONS

Away teams benefit enormously from a capable assistant.

Specialists carry mission equipment.

Security carries weapons.

Medical carries treatment.

The assistant carries everything else and somehow still has room for the mineral everyone claims is valuable.

An assistant may serve as:

- Porter.
- Lookout.
- Secondary miner.
- Radio operator.
- Rear guard.
- Casualty carrier.
- Oxygen custodian.
- Salvage handler.
- Spare pair of hands.
- Person repeatedly reminding the team that the submarine has recalled them.

Do not wander off.

Do not touch artifacts without permission.

Do not collect every object in a wreck simply because it can be picked up.

Do not announce that you have found something “very important” and then return carrying a chair.

Unless the submarine genuinely needs a chair.

In that case, the quartermaster may be forced to apologize.

THE ASSISTANT AND THE CLOWN QUESTION

At some point, an assistant may encounter clown clothing.

This may arrive through personal choice, local tradition, union ritual, station surplus, or an unknown process by which clown masks spontaneously appear wherever assistants gather.

Wearing clown attire does not automatically make an assistant incompetent.

It does, however, increase the burden of proof.

A clown carrying ammunition to the correct gun is useful.

A clown applying first aid is useful.

A clown welding the hull is useful, if mildly unsettling.

A clown honking repeatedly beside the captain during a docking maneuver is conducting an experiment in security response times.

Morale is valuable.

Timing is more valuable.

Learn the distinction between relieving tension and creating a new reason for tension.

WHEN ALL SPECIALIST ROLES ARE TAKEN

Sometimes you join a crew and discover that every specialist position already has someone assigned.

Good.

This means the vessel may actually possess enough qualified personnel to survive.

It does not mean you have no role.

The crew does not become complete merely because every uniform has been filled.

A submarine with one person in each specialty may still have:

- One engineer handling twenty junction boxes.
- One mechanic responsible for the entire hull.
- One doctor treating several casualties.
- One security officer defending multiple decks.
- One captain trying to navigate while everyone speaks at once.

The assistant gives these people breathing room.

You are not competing with the specialist.

You are expanding what the specialist can accomplish.

A second set of hands is not redundant when the first set is holding a welding tool and trying to keep the sea outside.

FINDING YOUR FUTURE SPECIALTY

Many excellent specialists begin as assistants.

The assistant role lets you observe the entire vessel before deciding where your aptitudes belong.

You may discover that you enjoy:

- Repairing machinery.
- Managing power.
- Treating injuries.
- Defending the crew.
- Navigating.
- Mining.
- Fabrication.
- Logistics.
- Away-team operations.
- Organizing everyone else while insisting you have no desire to become captain.

Take time to learn.

You do not need to select a specialist career during your first voyage.

Practice under supervision.

Ask questions.

Build familiarity.

Learn which alarms require immediate action and which alarms merely indicate that someone has turned off the oxygen generator intentionally.

When you feel prepared, speak with the relevant department head.

A mechanic will appreciate an assistant who already knows the pumps.

An engineer will appreciate one who understands junction boxes.

A doctor will appreciate one who knows that painkillers are not a universal cure.

Security will appreciate one who has mastered the advanced doctrine of not shooting through teammates.

THE VETERAN ASSISTANT

Some assistants never transfer into a specialist position.

This is not because they failed to develop.

It is because they developed in every direction.

A veteran assistant may know:

- Enough engineering to stabilize the reactor.
- Enough mechanics to repair the hull.
- Enough medicine to preserve a casualty.
- Enough security procedure to defend a compartment.
- Enough navigation to hold depth.
- Enough logistics to know why every department is lying about its supply consumption.

They may not be the best person aboard at any individual profession.

They may be the second-best person aboard at five of them.

That makes them exceptionally valuable.

When a specialist is incapacitated, the veteran assistant fills the gap.

When the vessel is short-handed, they move where needed.

When several departments fail at once, they understand enough of each to keep the submarine alive until the crisis becomes organized.

The veteran assistant is not the crew member who has failed to specialize.

They are the crew member who has turned adaptability into a profession.

THE ASSISTANT'S PRIVATE SUPERPOWER

Every assistant eventually develops one of several unusual abilities.

Some can locate any missing tool.

Some can carry an impossible number of items.

Some always arrive with a fresh oxygen tank.

Some know which compartment is flooding before the alarm sounds.

Some can calm an angry engineer with the simple phrase:

“I already brought the components.”

Some can cross the entire submarine during combat without being shot, bitten, drowned, or accidentally locked in ballast.

Some somehow know exactly when the captain is about to ask for a status report.

These are not talents listed in recruitment material.

They emerge through repeated exposure to crisis, poor storage discipline, and coworkers who habitually leave essential equipment in their pockets.

Cultivate your ability.

Use it responsibly.

If your ability is “discovering new ways to poison the crew,” consider cultivating another one.

FINAL RULE OF THE GENERAL CREWMAN

The assistant is not a consolation prize.

It is not the chair left vacant after the important jobs have been assigned.

It is not a declaration that you are unskilled, unnecessary, or waiting to be replaced.

It is the crew position with the broadest field of action and the greatest freedom to learn.

A new assistant contributes by listening, carrying, observing, and responding.

An experienced assistant contributes by anticipating, organizing, repairing, supporting, and filling gaps before they become disasters.

A veteran assistant can make the entire submarine operate more smoothly simply by being present, alert, and willing to go where the work is accumulating.

Bring the mechanic welding fuel.

Bring security ammunition.

Bring medical casualties.

Bring engineering components.

Bring the captain accurate information.

Bring the away team home.

Bring back the tools everyone else forgot.

And when every department insists it is too busy to solve the latest problem, step forward and ask the traditional question of the European general crewman:

“All right. What has broken now?”

You are not the least important member of the crew.

You are the person who makes the important members of the crew substantially less helpless.

Wear that distinction proudly.

The novelty hat is optional.

THE CARRYING OF ARMS

A firearm aboard a submarine is not a symbol of rank, authority, or personal importance. It is a tool entrusted to a crew member for the defence of the vessel and the lives aboard it.

Security personnel, armed divers, and away-team members must understand that possessing a weapon does not require its immediate use. The ability to fire is not the same as having a safe opportunity to fire.

A single crawler outside the hull may not justify a nuclear railgun shell. A crew member wearing a clown suit is not automatically a hostile combatant. An argument in the reactor compartment is rarely improved by launching a grenade into it. A mudraptor surrounded by fleeing crew requires a clear line of fire, not indiscriminate panic.

Before firing, identify:

1. The threat.
2. What lies between you and the threat.
3. What lies behind the threat.
4. Which weapon is appropriate.
5. Whether friendly personnel are within or near the line of fire.
6. Whether the ammunition may be required against a greater threat later.

The use of force should always be proportionate to the danger.

When dealing with disorder among the crew, begin with clear instructions. Use restraint, stun weapons, handcuffs, or other nonlethal measures when circumstances allow. Lethal force is reserved for immediate and serious threats to life, the vessel, or the mission.

Security officers are expected to protect the crew, not terrorize them.

They are not permitted to manufacture confrontations, demand personal obedience beyond their assigned authority, or punish imagined disrespect. They do not execute crew members because they are irritating. They do not fire blindly into crowded compartments. They do not empty weapons into incapacitated personnel merely to demonstrate certainty.

Trigger discipline preserves ammunition, prevents accidental casualties, and maintains the crew's trust.

A security officer feared by every person aboard is not maintaining order. They are creating the next emergency.

LINES OF FIRE

Every armed crew member must understand one rule before carrying a loaded weapon:

If a crew member is standing between you and the enemy, your line of fire is not clear.

It does not matter how confident you are.

It does not matter how close the enemy appears to be.

It does not matter whether the crew member in front of you is standing, crouching, crawling, fighting, retreating, or being dragged across the deck.

It does not matter whether you believe you can fire over their shoulder, past their helmet, between their legs, or a fraction of a metre above their head.

Do not fire through another crew member.

Weapons are rarely as accurate under combat conditions as their operators imagine. The submarine may be moving. The target may lunge. The shooter may stumble. Recoil may pull the barrel off target. Visibility may be poor. Water, smoke, blood, fire, damaged lighting, confined corridors, and panicked movement may obscure the exact position of both friend and foe.

A shot that appears perfectly aligned may strike the person you are supposedly attempting to protect.

This is especially important when a crew member is already engaged in close combat with a hostile life-form. If a mechanic, diver, security officer, or assistant is being bitten, clawed, or dragged by a creature, firing through their body does not improve their chances of survival.

It gives them an additional injury.

A GSW, or gunshot wound, creates another source of bleeding, organ damage, pain, blood loss, and incapacitation. If your crewmate is already being mauled by the forces of the deep, shooting them in the back of the head is not assistance.

It is friendly fire.

The rule is therefore simple:

Friendly in front. Enemy behind. Do not fire.

End of instruction.

THE CROUCHING FALLACY

Some armed personnel believe that a crouched teammate provides sufficient clearance to fire over them.

This belief has filled more medical wards than any European predator.

A crouched crew member is not cover, a weapon rest, or an acceptable firing aperture. They may stand without warning. They may be struck and thrown upward. They may climb a ladder, recoil from an attack, stumble over loose equipment, or move directly into the path of your shot.

The enemy may also move lower at the exact moment you fire.

The supposed gap above a crouching crew member is not a safe corridor for bullets, harpoons, shotgun pellets, grenades, explosives, or directed energy.

If a friendly crew member is crouching between you and the target, the line of fire remains obstructed.

Do not fire.

ELEVATED AND OFFSET TARGETS

An enemy being slightly above, below, or to one side of a friendly does not automatically create a clear shot.

Do not assume that a weapon will place every round precisely where the sights are pointed. A burst may climb. A shotgun may spread. A harpoon may deviate. A moving shooter may drag the barrel across a friendly silhouette. An explosive projectile may detonate against a railing, doorway, hatch frame, or the crew member themselves.

If any portion of a crewmate occupies the likely path of the projectile, the shot is unsafe.

A narrow angle is not a clear angle.

A partial opening is not a clear opening.

A hopeful shot is not a clear shot.

Wait, move, or choose another method.

REPOSITION BEFORE FIRING

When a friendly blocks your shot, your first responsibility is not to fire more carefully.

Your first responsibility is to establish a safer angle.

You may:

- Move to the side of the engagement.
- Fall back and cover a wider approach.
- Use another corridor, hatch, ladder, or compartment.
- Order the forward crew member to disengage or move aside.
- Allow the closer combatant to finish the engagement.
- Prepare to fire if the friendly is thrown clear.
- Cover additional enemies approaching from another direction.
- Provide medical, defensive, or logistical support instead of shooting.

Do not crowd directly behind a crew member who is already fighting. Doing so restricts their ability to retreat and places your weapon directly against their back.

Spread out where the environment permits. Establish overlapping fields of fire that do not overlap friendly bodies.

A firing position is useful only when it allows you to engage the enemy without first shooting through your own crew.

CROSSING A FIRING LINE

Crew members not currently firing must also respect the weapons around them.

Do not run in front of an armed security officer who is actively engaging a target. Do not step through a doorway while someone is firing through it. Do not climb a ladder into an established firing position without warning the shooter.

When crossing is necessary, communicate.

Use clear calls such as:

- "Crossing!"
- "Hold fire!"
- "Moving in front!"
- "Cease fire!"
- "Friendly entering!"
- "Clear behind me!"
- "Taking point!"

The shooter should acknowledge the warning and stop firing before the crew member crosses.

The person carrying the weapon remains responsible for the shot. However, every member of an armed team shares responsibility for maintaining clear and predictable movement.

Do not dart unpredictably through active firing lanes and then blame the shooter for failing to anticipate your path.

Likewise, a shooter may not excuse reckless fire by claiming that a friendly “should not have moved.”

Both parties must communicate.

BOARDING ACTIONS AND ASSAULTS

During an assault on an enemy vessel, hostile outpost, pirate position, wreck, ruin, or occupied compartment, the danger of friendly fire increases sharply.

Corridors are narrow. Doorways create bottlenecks. Visibility is limited. Teams frequently stack behind one another. Hostile personnel may retreat through friendly positions, and crew members may rush forward without announcing themselves.

An assault team must maintain order.

The crew member at the front controls the immediate engagement. Personnel behind them do not fire through them merely because they can see part of the enemy.

If the front member cannot safely engage, they should move, crouch behind actual cover, withdraw, or communicate that another shooter should take the angle.

The second crew member should cover an alternate direction rather than placing a weapon against the first crew member’s spine.

Explosive weapons require even greater restraint. Grenades, grenade launchers, explosive ammunition, nuclear ordnance, and volatile compounds should not be used in occupied compartments unless the threat clearly justifies the probable casualties and structural damage.

Before throwing or launching an explosive, consider:

- The blast radius.
- The confined nature of the compartment.
- Nearby crew members.
- The integrity of the hull.
- Fuel, oxygen, ammunition, or reactor systems nearby.
- Whether the explosive may rebound from a hatch or wall.
- Whether the enemy is close enough to friendly personnel that both will be caught in the detonation.

A successful assault does not leave the enemy dead, the assault team dead, and the objective compartment open to the European sea.

HOSTILE BOARDERS ABOARD YOUR OWN VESSEL

When hostile boarders or creatures enter the submarine, indiscriminate fire becomes especially dangerous.

Every missed shot may strike:

- A crew member.
- A reactor control.
- A junction box.
- A fuel rod.
- Medical storage.
- Ammunition.
- A window or weakened section of hull.
- A pump required to control flooding.
- A volatile chemical container.
- An oxygen tank.
- An explosive stored nearby.

The objective is not merely to kill the intruder. The objective is to kill or contain the intruder while preserving the vessel and its crew.

Position defenders so that they fire away from occupied compartments whenever possible. Use doorways and hatches to control the enemy's approach. Do not surround a hostile creature and then fire toward one another from opposite sides.

If two security officers are positioned on opposite sides of an enemy, neither has a safe line of fire.

One must move.

Never create a firing arrangement in which a missed shot continues directly into another crew member.

AUTOMATIC FIRE AND SHOTGUNS

Automatic weapons create a false sense that accuracy can be replaced with volume.

It cannot.

Firing a longer burst does not make an obstructed line of fire safer. It merely increases the number of opportunities to strike a friendly.

Use controlled bursts. Stop firing when a friendly crosses the sight line. Reassess after the target moves. Do not continue holding the trigger while sweeping the weapon across a crowded compartment.

Shotguns require particular care. The user must account for the entire spread, not merely the centre of the weapon's aim.

A friendly standing beside the target may still be struck.

A friendly crouching beneath the target may still be struck.

A friendly fighting the target at close range is almost certain to be struck.

Do not use a shotgun through a cluster of crew members and then express surprise when medical reports multiple pellet wounds.

The spread is part of the weapon.

You are responsible for all of it.

HARPOONS, SPEARGUNS, AND HEAVY WEAPONS

Harpoons and spears may appear slower and more predictable than firearms, but the same line-of-fire rules apply.

Do not fire a harpoon through a friendly diver. Do not launch explosive harpoons into a close-quarters struggle. Do not assume a diving suit will protect a crewmate from a projectile intended for a mudraptor.

Heavy weapons demand still greater caution.

Railguns, flak cannons, coilguns, chainguns, pulse lasers, discharge coils, and depth charges can kill personnel outside the vessel as readily as the creatures attacking them.

Before firing an external weapon, confirm the location of the diving party.

If divers are near the target, communicate with them before firing. If communication has been lost, assume they may still be present until their location is confirmed.

The operator of a submarine weapon must know not only what appears on the periscope, but also who has been sent outside and where they were expected to operate.

A hostile contact beside an unaccounted-for dive team is not automatically a clear target.

FIRE DISCIPLINE

Fire discipline means knowing when to shoot, when to stop, and when not to begin.

A disciplined shooter:

- Keeps the weapon pointed away from friendly personnel.
- Keeps their finger off the trigger until prepared to fire.

- Identifies the target before engaging.
- Checks the foreground and background.
- Stops firing when a friendly enters the line of fire.
- Uses controlled fire rather than panic.
- Repositions when the firing angle is unsafe.
- Warns others before using explosives or heavy weapons.
- Reports accidental friendly fire immediately.
- Does not conceal a gunshot wound they caused.
- Assists the injured once the threat is controlled.

If you accidentally shoot a crew member, report it.

State who was hit, where they were hit, and whether the threat remains active.

For example:

- “Friendly hit in engineering. One GSW to the torso. Medic required.”
- “Accidental discharge near the airlock. Diver wounded.”
- “Cease fire. Security is in the line.”
- “Shotgun pellets struck the mechanic. Threat contained.”

Do not continue firing while pretending nothing happened. Do not blame the casualty before medical personnel have even arrived. Do not allow embarrassment to delay treatment.

A concealed mistake becomes negligence.

An immediately reported mistake can still be managed.

FINAL RULE OF THE FIRING LINE

The officer, diver, or crew member holding the weapon is responsible for every projectile leaving it.

The enemy does not excuse the shot.

Confusion does not excuse the shot.

Poor visibility does not excuse the shot.

The urgency of the battle does not excuse the shot.

If the line of fire is occupied by a friendly, do not fire.

If you are uncertain whether the line of fire is occupied, do not fire.

If you believe you can probably shoot over, under, around, or past a friendly, do not fire.

Move.

Wait.

Communicate.

Change weapons.

Change angles.

Help in another way.

Your crewmate already has enough to contend with. They do not need to survive claws, teeth, bullets, pressure, and the darkness of Europa only to be killed by the person standing behind them.

COMMUNICATION ABOARD SHIP

A submarine survives on accurate information delivered at the proper time.

The crew does not need a continuous account of every thought passing through your mind. It does need to know that the ballast tank has breached, the reactor is overheating, a mudraptor has entered engineering, the medic is incapacitated, or a diving party has not returned.

Mission-critical communication should be concise.

State the location, the problem, and what assistance is required.

Examples:

- "Flooding in aft ballast. One mechanic responding."
- "Fire in engineering. Electrical panel involved."
- "Two crawlers on the upper hull."
- "Medic required in the reactor compartment."
- "Dive team returning with one casualty."
- "Railgun ammunition low."
- "Unknown contact approaching from below."

Avoid speaking over the captain during navigation, over security during a boarding action, or over a medic coordinating casualty treatment. During an emergency, unnecessary radio traffic can conceal the one report that matters.

This does not mean crew members are forbidden from speaking, making suggestions, or maintaining morale. A silent vessel can become as dangerous as a chaotic one. Conversation is welcome when circumstances permit.

The important distinction is between communication and interference.

Report what the crew needs to know. Allow others to answer. Keep arguments off the main channel. Do not demand the captain's immediate attention for personal inconvenience while the vessel is sinking.

You are part of the crew, not the sole subject of the voyage.

THE MEDICAL OFFICER'S CHARGE

The medical officer's duty is to preserve the health, mobility, and survival of the crew.

It is not to distribute narcotics to anyone who wanders into the infirmary announcing, "I hurt."

It is not to empty every ampoule of morphine, fentanyl, antibiotics, anti-radiation medication, combat stimulants, and experimental compounds into the nearest crew member merely because the vessel has encountered mild turbulence.

It is certainly not to become the dealer.

The title of medical officer carries considerable authority aboard a submarine because the crew must trust that the person standing over them with a syringe has some understanding of what is inside it. On Europa, that trust is frequently granted long before the medical officer has acquired anything resembling a terrestrial medical licence.

Many submarine medics are former field orderlies, emergency technicians, pharmaceutical workers, research assistants, military corpsmen, self-taught practitioners, or people who were handed a medical uniform because they were the only applicant who could distinguish blood from hydraulic fluid.

Formal licensing may therefore be uncommon.

Basic competence is not optional.

A crew member who agrees to serve as medical officer is expected to understand the vessel's medical systems, recognize common injuries, use diagnostic equipment, administer appropriate treatment, avoid preventable overdoses, preserve medical supplies, and establish priorities when several casualties require attention at once.

A medic does not need to cure every injury instantly.

They do need to avoid becoming the leading cause of death aboard the vessel.

THE PURPOSE OF MEDICAL TREATMENT

Medical treatment exists to restore a crew member to useful health, prevent an injury from worsening, and keep the casualty alive until more complete treatment is available.

These are not always the same objective.

A minor bruise may require no treatment beyond time and caution. Moderate bleeding may require bandaging before pain relief is considered. A crew member suffering oxygen deprivation requires breathable air before narcotics. A patient with catastrophic blood loss requires blood replacement more urgently than they require reassurance.

The medical officer must determine what is wrong before reaching for medication.

That evaluation does not need to become an hour-long consultation while the submarine fills with water. It should, however, involve more than asking, "Does it hurt?" and injecting the patient with the first substance found in the cabinet.

At minimum, determine:

- What injured the patient.
- Where the injury is located.
- Whether the patient is bleeding.
- Whether significant blood has already been lost.
- Whether the patient can breathe.
- Whether they are conscious and responsive.
- Whether they have been poisoned, infected, burned, irradiated, or exposed to pressure.
- Whether they have already taken medication.
- Whether another casualty requires more urgent attention.

The medic's purpose is not to eliminate every discomfort immediately. It is to identify which conditions are dangerous, which are becoming dangerous, and which can safely be left to recover without consuming limited supplies.

COMMON CONDITIONS OF THE SUBMARINER'S TRADE

A competent medical officer should be able to recognize and respond to the injuries most frequently encountered aboard a European submarine.

Lacerations and Bleeding

Cuts and lacerations may result from creature attacks, broken machinery, shattered structures, knives, harpoons, or personnel attempting to repair electrical equipment while holding entirely inappropriate tools.

The immediate concern is uncontrolled bleeding.

Stop the bleeding before treating the patient's complaints about pain. A conscious crew member with a painful wound can wait several seconds for analgesia. A crew member rapidly emptying their circulatory system onto the deck cannot.

Bandages and other wound treatments should be used according to the severity of the injury. Do not consume advanced medical supplies on scratches that could be managed with simpler treatment.

Once bleeding has been controlled, assess blood loss and any underlying injury.

Gunshot Wounds

A gunshot wound, commonly reported over the radio as a GSW, may involve bleeding, internal trauma, organ damage, pain, loss of mobility, or rapid unconsciousness.

The fact that security describes the shooting as accidental does not make the injury less severe.

Determine the number and location of wounds. Control bleeding. Evaluate blood loss. Stabilize the patient before beginning an argument about who crossed whose firing line.

If an armed crew member has shot a teammate, the medic requires useful information:

- What weapon was used.
- How many times it was fired.
- Where the patient was struck.
- Whether the area is now secure.
- Whether additional casualties are incoming.

“Something happened in engineering” is not an adequate medical report.

Blood Loss

Blood loss is a condition in its own right and must not be mistaken for ordinary pain.

A patient may appear pale, weak, unsteady, confused, or close to unconsciousness. Closing the wound prevents further loss, but it does not replace what has already been lost.

A medic who bandages a heavily exsanguinated patient and immediately sends them back to work has completed only half the treatment.

Blood products and replacement fluids must be preserved for patients who genuinely require them. They should not be used as refreshments, demonstrations, practical jokes, or ingredients in experiments conducted without the captain’s knowledge.

Blunt-Force Trauma

The submarine is filled with ladders, hatches, machinery, narrow access ways, steep companionways, and crew members who believe descending headfirst is faster.

As a result, the infirmary will receive a steady procession of people who have fallen down ladders, collided with doors, been struck by loose equipment, or injured themselves during abrupt changes in depth and motion.

Not every bruise requires narcotics.

Minor blunt-force injuries can frequently be observed and allowed to recover naturally. A crew member suffering a small amount of pain after falling down a ladder may be advised to rest, work more carefully, and—in the traditional medical terminology of the European fleet—sleep it off.

More serious trauma requires attention when it causes substantial pain, impaired movement, unconsciousness, internal injury, or a rapidly worsening condition.

The medical officer must distinguish discomfort from disability.

Burns

Burns may result from fire, reactor accidents, electrical discharge, welding incidents, chemical exposure, or a mechanic attempting to extinguish flames by standing inside them.

Remove the casualty from the source of injury. Extinguish burning clothing. Treat the burn with appropriate supplies and consider whether the patient has also suffered smoke inhalation, oxygen deprivation, or electrical injury.

Do not inject a heavily burned casualty with an impressive collection of medication merely because the injuries look dramatic. Treat the actual conditions present.

Oxygen Deprivation

A patient suffering from oxygen deprivation requires oxygen.

This principle appears obvious, but it has repeatedly escaped medical personnel who attempt to treat unconscious divers with pain medication while leaving an empty oxygen tank attached to the patient's suit.

Check the airway, the surrounding atmosphere, the diving suit, and the oxygen supply. Move the patient into a breathable compartment when possible.

Medication cannot compensate for the continued absence of oxygen.

Pressure and Diving Injuries

Personnel exposed to the European sea without adequate protection may suffer catastrophic pressure trauma. These casualties require immediate recovery into a safe compartment and rapid treatment.

The medical officer should know when a diving team has departed, what hazards they expect to encounter, and which supplies may be required upon their return.

A medic who learns that an away team exists only when three unconscious divers are deposited on the infirmary floor has not been properly monitoring the operation.

Poisoning

Europa offers a remarkable variety of substances capable of killing a human being.

Some are produced by local life. Others are carried aboard deliberately. Still others were fabricated by the medical department after someone discovered that the chemical synthesizer would permit it.

The medical officer should understand the common effects of the poisons, venoms, paralytics, and toxins carried or encountered by the crew. They must also know which antidotes correspond to which agents.

An antidote is not universally effective merely because the container contains the word "antidote."

When poisoning is suspected, determine what substance may have been involved and how long ago exposure occurred. Administer the appropriate countermeasure and observe the patient for continuing deterioration.

Do not respond to an unknown toxin by injecting every antidote aboard the vessel in alphabetical order.

Infection and Parasites

Any crew member returning from a wreck, ruin, cave, nest, or flooded compartment may have encountered infection.

Unexplained difficulty breathing, abnormal wounds, unusual movement, progressive weakness, behavioral changes, or visible signs of parasitic exposure should be investigated immediately.

The medical officer must balance treatment with containment. A suspected infection may endanger more than the patient.

Inform the captain when quarantine is necessary. Do not conceal a serious communicable condition because the infected crew member is a friend, senior officer, or particularly persuasive clown.

Radiation and Environmental Exposure

Depending on the route, mission, equipment, and condition of the vessel, the crew may encounter radiation or other environmental hazards.

Anti-radiation treatment is not a recreational stimulant and should not be issued merely because someone stood near the reactor and became nervous.

Confirm meaningful exposure before consuming scarce treatment.

The engineer's continued habit of sleeping beside the reactor does not automatically constitute a medical emergency, though it may warrant a separate discussion.

Pain, Shock, and Psychological Effects

Pain is medically relevant, but pain alone does not determine treatment priority.

A shouting patient with a minor wound may appear more urgent than a silent patient who is bleeding internally, losing oxygen, entering shock, or becoming unconscious. The medical officer must treat severity rather than volume.

Shock, confusion, panic, psychosis, and other psychological conditions can also affect crew safety. A patient may become irrational due to injury, poisoning, oxygen deprivation, medication, fear, infection, or an underlying psychological condition.

Do not assume every distressed crew member is merely being difficult.

Do not assume every difficult crew member requires sedation.

Assess the situation, reduce immediate danger, and use restraint or medication only when justified.

“I HURT” IS A SYMPTOM, NOT A DIAGNOSIS

Crew members will frequently enter medical and announce that they are injured without providing any further information.

The medical officer is expected to evaluate them, not automatically medicate them.

Ask where they are hurt and what happened. Use the health scanner or other diagnostic tools. Determine whether the condition is serious enough to justify treatment.

A crew member who has suffered a minor bruise, a small laceration, or ordinary soreness may not require narcotics. They may need a bandage, observation, a brief rest, or nothing at all.

This is not neglect.

It is responsible resource management.

Morphine is not a reward for reporting to medical. Fentanyl is not a morale ration. Combat stimulants are not refreshments. Sedatives are not tools for making annoying crew members quieter.

If the patient does not medically require the drug, do not administer it.

The medic who responds to every complaint with narcotics will soon discover that the crew has developed an extraordinary number of mysterious aches immediately before every mission.

NARCOTICS AND CONTROLLED MEDICATION

The submarine's medical cabinet may contain a staggering variety of narcotics, stimulants, sedatives, poisons, antidotes, antibiotics, anti-radiation treatments, and experimental compounds.

Access to these substances is granted because the medical officer may need them to save lives.

It is not granted so the infirmary can become the liveliest social compartment on the vessel.

The medical officer must control access to high-risk medication. Do not distribute narcotics merely because a crew member asks. Do not leave dangerous drugs scattered across the deck. Do not hand a crate of morphine to an assistant because they promise to “be responsible.”

A medication can be appropriate in one dose and lethal in another.

Before administering a controlled drug, consider:

- The patient's condition.
- The seriousness of their pain.
- Other drugs already administered.
- The likelihood of overdose.
- The possibility of addiction or withdrawal.
- The supply remaining aboard.
- Whether a safer treatment is available.

Do not repeatedly medicate a patient without checking what they have already received. In an emergency involving several medics, communicate doses clearly.

A patient should not receive three separate injections from three separate medical personnel because none of them spoke to one another.

Mark treated casualties when practical. Report significant medication over the radio or through established medical channels.

“Patient stabilized with morphine” is useful.

“I gave them something” is not.

OVERDOSE IS NOT AN EXPERIMENT

There will occasionally be a crew member who wishes to determine how many doses of morphine, fentanyl, stimulants, or other medication a human body can tolerate.

The medical officer's role is to prevent this experiment, not supervise it.

The answer is not medically useful. It is operationally expensive and frequently fatal.

Encouraging the crew to inject themselves for amusement consumes supplies that may later be required during boarding actions, salvage operations, mining expeditions, nest clearances, ruin exploration, or catastrophic hull failure.

The captain will not appreciate being informed at the next outpost that another crate of morphine is required when the crew suffered no serious injuries during the previous voyage.

The following explanation will not improve matters:

“Jerry in engineering wanted to know how many injections it would take to kill someone, and the other half of the crew volunteered to help establish the answer.”

The answer, for procurement purposes, will invariably be:

Every dose we had.

A medical officer who encourages this degree of waste should expect to be voluntold into the next several sea-poppy harvesting expeditions. They may spend a considerable portion of subsequent voyages scraping aquatic poppies from cave walls under the supervision of an irritated security officer.

Medical supplies are purchased to keep the crew alive.

Do not use them to establish exactly how difficult that task can be made.

POISONS, VENOMS, AND THEIR ANTIDOTES

The medical department may possess substances that have legitimate medical, research, security, or mission-specific uses but are also capable of causing paralysis, organ failure, unconsciousness, psychosis, or death.

These materials must be controlled.

Do not carry poisons casually in personal clothing. Do not leave them beside ordinary medication. Do not replace legitimate ampoules with dangerous compounds as a joke. Do not inject unconscious patients with experimental chemicals unless circumstances justify the risk and the procedure has been authorized.

Labels should remain visible. Dangerous substances should be stored separately from routine treatment whenever the vessel's layout permits.

The medical officer should maintain sufficient antidotes to respond to the toxins most likely to be encountered. They should not consume the entire antidote supply treating imagined exposure or experimenting on volunteers.

A poison and its antidote are not a matched recreational set.

TRIAGE DURING MASS-CASUALTY EVENTS

When several crew members are injured simultaneously, the medical officer must establish priorities.

The loudest patient is not necessarily the most critical.

Treat first those who will die without immediate assistance and who can still be saved through timely intervention. Then treat those whose conditions may deteriorate. Minor injuries can wait.

A practical order of attention is:

1. Patients who cannot breathe or are suffering severe oxygen deprivation.
2. Patients with uncontrolled bleeding or catastrophic blood loss.
3. Patients who are unconscious, rapidly deteriorating, or suffering life-threatening poisoning.
4. Patients with severe burns, major trauma, or dangerous infection.
5. Patients with moderate injuries that impair their ability to function.
6. Patients with minor injuries and manageable pain.
7. Crew members requesting narcotics because they bumped their knee while climbing a ladder.

Triage may require directing uninjured personnel to assist.

A mechanic can apply a bandage. A security officer can carry a casualty. An assistant can retrieve blood packs or oxygen. An engineer can help move patients out of a flooded compartment.

The medical officer should coordinate these efforts rather than attempting to perform every task alone.

During a major emergency, give direct instructions:

- "Apply pressure to that wound."
- "Bring two blood packs."
- "Move this patient into oxygen."
- "Carry the unconscious diver to medical."
- "Do not administer anything else."
- "Security, restrain the infected patient."
- "Captain, medical requires evacuation."

Clear medical command saves lives.

Frantic rummaging through cabinets does not.

PREPARING AWAY TEAMS

The medical officer's responsibility extends beyond the infirmary.

Away missions expose personnel to hostile creatures, gunfire, pressure, drowning, infection, venom, environmental hazards, and the perennial danger of someone forgetting to bring enough oxygen.

Before a salvage, mining, exploration, assault, or recovery operation, the medical officer should help prepare an appropriate field kit.

The kit should reflect the actual mission.

A short mining excursion near the submarine does not require the entire medical cabinet. An assault on a hostile installation should not depart with one bandage and a positive attitude.

An away-team medical kit may include:

- Basic wound treatment.
- Supplies for controlling serious bleeding.
- Blood replacement material when justified.
- Appropriate pain relief.
- Antidotes relevant to the expected threat.
- Treatment for burns or environmental exposure.
- Additional oxygen.
- Emergency stimulants or revival equipment.
- Tools required to transport or stabilize casualties.

Distribute supplies sensibly. Do not place the entire medical reserve on one diver who may be the first person incapacitated.

The medical officer should know who is carrying the field supplies and whether a trained medic is accompanying the mission. When possible, retain enough material aboard to treat casualties caused by a simultaneous attack on the submarine.

Europa has a habit of creating two emergencies whenever the crew prepares for one.

MEDICAL READINESS ABOARD THE VESSEL

A good medical officer does not wait for someone to begin dying before checking whether the department is functional.

Before departure, inspect medical storage.

Confirm that:

- Essential medicines are present.
- Blood and emergency supplies are available.
- Dangerous drugs are secured.
- Diagnostic equipment is operational.
- Diving medical kits are prepared.
- Empty or depleted containers are identified.
- Supplies are stored where they can be found quickly.
- Previous patients have not wandered away carrying half the pharmacy.

During quieter periods, fabricate or organize commonly required treatments. Monitor the consumption of scarce materials. Inform the captain before the submarine reaches a crisis.

“Medical is low on blood packs and broad-spectrum antibiotics” is useful before the next mission is selected.

“We have nothing left” is less useful after the crew has been boarded.

THE MEDICAL OFFICER IS NOT EXEMPT FROM ACCOUNTABILITY

Medical authority does not permit experimentation without consent, poisoning crew members for entertainment, withholding care as punishment, or using sedation to settle arguments.

Nor does it excuse incompetence that the officer refuses to correct.

Every medical officer will make mistakes. European medicine is performed in moving vessels under poor lighting while alarms sound and creatures attempt to enter the room. Perfection is not expected.

Learning is expected.

When an error occurs, report it honestly. If the wrong medication was given, say what was administered and how much. If supplies were lost, tell the captain. If a patient may have been exposed to infection, do not conceal it.

A medic who admits a mistake gives the crew an opportunity to correct it.

A medic who hides a mistake converts it into a second emergency.

FINAL DUTY OF THE MEDICAL OFFICER

The crew does not require the medical officer to be a miracle worker.

It requires them to be dependable.

Learn the common afflictions.

Understand what the medicines aboard the vessel actually do.

Treat dangerous conditions promptly.

Allow minor injuries to recover without consuming strategic reserves.

Control narcotics, poisons, and antidotes.

Prepare away teams for the hazards they are likely to face.

Maintain enough supplies to survive the voyage that has not yet gone wrong.

Above all, remember that medicine is not measured by how many injections have been administered. It is measured by how many crew members remain alive, functional, and capable of returning to their stations.

You are not the dealer.

You are the medical officer.

Please conduct yourself in a manner that permits the crew to recognize the difference.

GOING ASHORE

Station Leave, Resupply, and Preparation Between Voyages

When a submarine docks at an inhabited station, members of the crew may be granted leave to go ashore.

“Going ashore” is the traditional expression, although there is very little shore beneath Europa’s ice and what passes for dry land is generally a pressurized corridor bolted into a cavern wall. The meaning remains the same: the vessel is secure, the docking connection has been established, and the crew may temporarily leave the submarine to enter the station.

Unless otherwise ordered by the captain or station authorities, crew members on leave are free to explore the station, speak with its inhabitants, visit merchants, seek local employment, purchase personal equipment, and spend their own salary as they see fit.

Your wages belong to you.

You may save them. You may purchase tools, clothing, weapons where permitted, medical equipment, personal luxuries, questionable food, decorative objects, or enough ethanol to make your next financial review deeply embarrassing.

The captain is not ordinarily responsible for supervising every personal purchase made by the crew. Neither is the captain required to finance those purchases from the vessel’s operational budget.

A distinction must always be maintained between:

- Personal salary, belonging to the individual crew member.
- Ship funds, reserved for the operation, maintenance, and continued survival of the submarine.

Do not ask the captain to purchase a recreational item from ship funds simply because you want one.

A novelty water gun, ceremonial hat, musical instrument, crate of alcohol, decorative pet, or exotic chemical may be entertaining. It is not automatically a legitimate vessel expense.

The captain may choose to authorize such purchases when the submarine is profitable and adequately supplied. They may even decide that crew morale justifies buying enough ethanol to leave engineering drunk as a monkey’s uncle.

That is the captain’s decision.

It should not become the subject of repeated demands while the command staff is calculating whether the submarine possesses enough fuel and ammunition to survive the next crossing.

PORT IS NOT WASTED TIME

Crew members are often eager to leave a station as quickly as possible.

This is understandable. A submarine earns little while sitting motionless in a docking berth. No cargo is being delivered. No wreck is being salvaged. No mineral deposit is being harvested. No bounty is being collected. The vessel is making no visible progress toward its next objective.

To use the language occasionally heard from less patient crew members, the voyage is not “moving forward.”

This view mistakes movement for progress.

Time spent at a station is frequently as valuable as time spent underway.

While docked, the submarine's immediate power and pressure requirements are supported by the station's infrastructure. The reactor can be taken out of its ordinary operating cycle. Batteries can be restored. Damaged systems can be examined without the vessel actively descending through hostile waters. Supplies can be counted in relative safety. Departments can compare what was consumed during the previous mission against what will be required during the next.

It may appear that nothing is happening.

In reality, the next voyage is being made possible.

A port call is often the crew's only reliable period of respite between emergencies. It is a time to rest, reorganize, replace missing personnel, purchase supplies, review damage, fabricate equipment, examine new mission opportunities, and determine what dangers lie ahead.

Rushing the submarine away from the station without proper preparation may save several minutes at the dock and cost the crew an entire voyage beneath the ice.

THE LOGISTICAL CONFERENCE

During a port call, the captain must coordinate with the vessel's departments.

This may involve the engineer, mechanic, medical officer, security personnel, gunnery crews, quartermaster, logistics officer, or whichever unfortunate crew member has inherited responsibility for keeping track of the ship's stores.

These conversations are not idle chatter.

They are how the captain determines:

- What was expended during the previous voyage.
- What was damaged or lost.
- What remains aboard.
- What must be purchased immediately.
- What can be fabricated from existing materials.
- What may be required by the next mission.
- Which supplies must be reserved for a longer crossing.
- Whether the vessel can afford discretionary purchases.
- Whether the selected mission is practical with the resources available.

Interrupting this process with repeated requests for toys, alcohol, recreational chemicals, personal equipment, or nonessential modifications does not make departure happen sooner.

It makes departure happen later.

Every time the captain must stop reviewing fuel reserves to answer whether the crew may purchase a super-soaker, the actual work of preparing the vessel is delayed.

Every time the engineer must leave a discussion about reactor endurance to explain why the submarine does not need twelve decorative lamps, the logistics process slows.

Every time the medical officer is interrupted while reporting shortages because someone wishes to know whether station alcohol is cheaper by the bottle or the crate, a necessary detail may be forgotten.

Nonessential requests can wait until the vessel's operational needs have been established.

The quickest way to leave port is to help complete the work required before departure.

DEPARTMENTAL INVENTORIES

Every department should inspect its own equipment before wandering away from the submarine.

Do not assume someone else knows what you used.

Do not assume the captain has personally counted every shell, bandage, fuel rod, oxygen tank, welding cylinder, diving suit, and screwdriver aboard the vessel.

The captain directs the process.

The departments provide the information.

Before going ashore for personal business, determine what your section consumed, lost, damaged, or requires.

Report shortages while the submarine is still docked beside merchants capable of correcting them.

Discovering a critical shortage after departure is not merely inconvenient. It means the crew had an opportunity to solve the problem and failed to use it.

SECURITY AND GUNNERY REPORTS

Security personnel and weapon operators should inspect the vessel's ammunition reserves after every substantial engagement.

Count ammunition by type, not merely by container.

Report:

- Empty or partially depleted ammunition boxes.
- Railgun shells expended.
- Coilgun, chaingun, flak, or pulse ammunition consumed.
- Nuclear or explosive ammunition remaining.
- Handheld ammunition used by security personnel.
- Lost or damaged weapons.
- Stun weapons, restraints, armor, or diving equipment requiring replacement.
- Specialized ammunition likely to be needed for the next mission.

A report stating "we still have some ammunition" is not sufficient.

The captain and engineering department need to know whether "some" means six full boxes or fourteen rounds rattling around in the bottom of an otherwise empty container.

Ammunition types are not interchangeable in purpose.

The rounds effective against small, fast-moving creatures may not be appropriate against armored targets. Weapons used for defending the submarine may require different supplies from those carried by an assault team boarding an enemy installation.

When the captain returns with proposed mission objectives, security should consider the expected threat.

If the next route is likely to involve pirate activity, hostile outposts, large creatures, swarms, nest clearance, or dangerous wreck exploration, say what ammunition and equipment will be required.

Do not remain silent at port and then announce halfway through the battle that the weapon best suited to the enemy has been empty since the previous voyage.

MEDICAL REPORTS

The medical officer must inventory the infirmary before departure.

This includes more than glancing into one cabinet and concluding that several colorful ampoules remain.

Report shortages of:

- Morphine and other necessary analgesics.
- Bandages and wound-treatment supplies.
- Blood packs and replacement fluids.
- Antidotes.
- Antibiotics.
- Anti-radiation medicine.
- Emergency stimulants.
- Sedatives.
- Treatment for burns and environmental injuries.
- Supplies needed for away-team medical kits.
- Any specialist medicine required by the anticipated mission.

The captain cannot purchase what the medical officer does not report.

If the submarine leaves a well-stocked station and the medic later announces that no morphine remains aboard, the captain is entitled to ask several pointed questions.

The vessel was just in port.

The merchants had medical supplies.

The ship possessed funds.

The opportunity existed.

Why was nothing said?

A shortage concealed through inattention is functionally little different from a shortage created through waste. In both cases, a wounded crew member may suffer because the medical department failed to perform a basic responsibility.

Medical personnel should also explain unusual levels of consumption.

If the infirmary began the last voyage with a full narcotics cabinet and returned with nothing despite suffering no serious casualties, "Jerry from engineering encouraged everyone to see how much morphine it would take to kill them" will not improve the captain's disposition.

Inventory is not merely counting.

It is accountability.

ENGINEERING AND MECHANICAL REPORTS

Engineers and mechanics should inspect the submarine's operational reserves before departure.

Critical concerns include:

- Reactor fuel rods.
- Welding fuel.
- Oxygen tanks.
- Diving-suit oxygen supplies.
- Portable batteries.
- Electrical components.
- Mechanical repair materials.
- Replacement tools.
- Fabrication resources.
- Hull-repair supplies.
- Fire extinguishers.
- Damaged pumps, engines, batteries, junction boxes, or docking systems.
- Materials required to manufacture ammunition or medicine underway.

Saying, "There is a fuel rod in the reactor," is not a complete fuel report.

The relevant questions are:

- How much life remains in that rod?
- Are replacement rods aboard?
- How much power will the next route require?
- Will the submarine be operating high-load systems?
- Can additional fuel be fabricated?
- Is the next outpost close enough to depend upon immediate resupply?
- What happens if the vessel must reverse course or remain active longer than expected?

The same reasoning applies to welding fuel.

One partially filled welding tank may be sufficient while the submarine is docked and intact. It will not remain sufficient after multiple hull breaches, a prolonged creature attack, and an away team returning with three empty tanks.

Inspect reserves according to the likely demands of the next voyage, not according to the calm conditions present at the station.

PERSONAL PURCHASES AND SHIPBOARD READINESS

Crew members may spend their salary as they please, but personal purchases must not interfere with the vessel.

Do not obstruct corridors with crates of personal goods. Do not store unstable chemicals beside reactor controls. Do not place recreational items in emergency cabinets. Do not fill the diving locker with alcohol and then complain that no room remains for diving suits.

Weapons purchased ashore must be handled according to the captain's rules and the station's laws. Dangerous chemicals, poisons, explosives, and experimental compounds should be declared before being brought aboard.

The phrase "I bought it with my own money" does not make every object safe to store beside the ammunition magazine.

Likewise, shore leave does not excuse a crew member from being fit for duty when the submarine departs.

If the captain has authorized drinking, drink with the understanding that the vessel may still need to leave port. Emergencies, mission opportunities, station unrest, hostile action, or docking restrictions may shorten leave unexpectedly.

A crew member who returns aboard so intoxicated that they cannot climb a ladder, operate their equipment, or distinguish the airlock from the ballast tank has converted a personal purchase into a shipboard problem.

Freedom ashore comes with the responsibility to return capable of surviving the departure.

REMAIN AVAILABLE FOR RECALL

Before leaving the submarine, know whether the captain has established a departure time.

Maintain radio contact when possible. Listen for recall announcements. Do not enter restricted station areas, distant ruins, or inaccessible tunnels without informing someone where you are going.

The captain should not be required to search every bar, merchant stall, maintenance corridor, and prison cell on the station because one crew member vanished without explanation.

When recalled:

1. Finish the transaction you are conducting.
2. Return to the submarine.
3. Report aboard.

4. Store your purchases safely.
5. Return to your assigned station or assist with final preparations.

If you expect a delay, report it immediately.

“Returning from medical merchant.”

“Held at station security.”

“Completing ammunition delivery.”

“Two minutes from docking bay.”

These are useful reports.

Silence is not.

The rest of the crew should not be held indefinitely because one submariner is still attempting to negotiate a discount on a decorative hat.

THE EXTENDED VOYAGE

As a submarine travels toward the less settled edges of European society, stations become more distant, less reliable, and more poorly stocked.

The voyage between ports may extend across many kilometres of hostile water. The next station may lack the supplies available at the current one. It may be abandoned, damaged, quarantined, occupied by hostile forces, or unwilling to trade.

A vessel preparing for such a crossing must think beyond the next few minutes.

Extended voyages require reserves.

This may mean carrying:

- Multiple reactor fuel rods.
- Additional welding and oxygen tanks.
- Spare ammunition for every installed weapon.
- Materials for fabricating additional supplies.
- Reserve medical treatment.
- Away-team provisions.
- Replacement batteries and tools.
- Backup diving equipment.
- Supplies for unexpected return travel.
- Enough material to survive more than one major engagement.

The correct amount depends upon the submarine, route, mission, crew size, installed equipment, and expected danger.

There is no universal answer.

There is, however, one reliably incorrect answer:

“We have one fuel rod in the reactor. We cannot possibly need another.”

A reserve is not wasted simply because it was not consumed.

A fuel rod that returns unused has still served its purpose by ensuring that the submarine was not stranded.

A medical kit that remains unopened was not unnecessary.

A box of ammunition that was not fired still provided the vessel with the ability to respond to an attack.

Preparation should be measured against what could reasonably happen, not merely against what happened during the easiest previous voyage.

HOW TO CONTRIBUTE PRODUCTIVELY IN PORT

Crew members wishing to accelerate departure should not repeatedly ask when the submarine is leaving.

They should help make the submarine ready to leave.

Useful work includes:

- Counting departmental supplies.
- Reporting shortages.
- Refilling oxygen and welding tanks.
- Recharging batteries.
- Restocking diving equipment.
- Recovering loose items from the deck.
- Sorting ammunition.
- Repairing damaged systems.
- Deconstructing unwanted salvage.
- Fabricating commonly used supplies.
- Preparing away-team kits.
- Moving purchased supplies aboard.
- Updating the captain on departmental readiness.
- Learning the likely requirements of the next mission.

A concise departmental report is especially valuable.

For example:

“Engineering ready. Two full fuel rods in reserve, six welding tanks filled, one junction box still damaged, and we need more electrical components before the long crossing.”

Or:

“Security ready. Coilgun ammunition is adequate, but we have one railgun shell remaining and no explosive rounds. Recommend purchasing shells before the nest-clearing mission.”

Or:

“Medical ready except for morphine and blood packs. Away-team kit is prepared, but we need replacements before departure.”

These reports permit the captain to make decisions.

“Can we go yet?” does not.

FINAL RULE OF SHORE LEAVE

A station call belongs to the whole crew.

It is a period of rest, commerce, recreation, repair, planning, and preparation. Crew members should be permitted to enjoy it, spend their wages, explore the station, and recover from the strain of the previous mission.

They should also recognize that the submarine cannot leave safely until its needs have been identified and met.

Enjoy your leave.

Buy what you please with your own salary.

Drink when permitted.

Explore within reason.

But before the docking clamps release, make certain your department has reported what it used, what it lost, what it lacks, and what it expects to need.

There are few sounds more frustrating to a captain than hearing, ten minutes after departure:

“Captain, did anyone buy fuel?”

Europa provides enough surprises.

The crew should not manufacture additional ones at the supply counter.

KEEP THE DECKS CLEAR

Loose objects are a hazard.

They obstruct movement, conceal damage, make emergency supplies difficult to locate, create noise, and may be swept out of the vessel during a hull breach. A tool abandoned in a corridor is of no use when the engineer searching for it is three compartments away and taking on water.

Return equipment to appropriate storage.

Place diving equipment near the airlock. Return medical supplies to medical storage. Store ammunition near the weapon it serves, unless the vessel's officers have established another arrangement. Refill tanks and place them with the full supplies. Separate empty, damaged, or contaminated equipment when possible.

Reusable equipment should not be discarded casually.

Harpoons can be recovered. Oxygen and welding tanks can be refilled. Empty containers, spent materials, and unwanted items may be recycled or deconstructed. Tools dropped during an emergency should be collected once the immediate danger has passed.

Bodies must also be handled with care.

The dead may carry identification, equipment, evidence of infection, or valuable supplies. More importantly, they were members of the crew. When conditions permit, recover and secure the remains. Follow the captain's instructions regarding burial, quarantine, storage, or transfer at the next outpost.

Dismembering a fallen crew member and dropping the remains into the ballast tank is not a recognized funerary custom.

A well-kept submarine is not merely pleasing to look at. It is faster to repair, easier to defend, safer to evacuate, and harder to lose in confusion.

Order aboard ship is a survival mechanism.

PROFESSIONAL CONDUCT

Life beneath the ice is stressful.

Crew members will make mistakes. Orders will be misunderstood. Equipment will fail at inconvenient moments. People will become frightened, tired, impatient, and angry. A submariner is not expected to remain cheerful through every disaster.

They are expected to remain civil.

Abuse, harassment, deliberate humiliation, threats, and sustained personal hostility have no place aboard a functioning vessel. They damage concentration, discourage necessary communication, and often escalate minor disagreements into violence.

Correct mistakes without degrading the person who made them.

A clear instruction is more useful than an insult. A concise explanation is more effective than a public tirade. If someone is inexperienced, teach them when circumstances permit. If they are unwilling to learn or repeatedly endanger the vessel, report the matter through the proper chain of command.

There is a difference between incompetence and sabotage. There is also a difference between harmless foolishness and conduct that threatens the ship.

Not every joke requires punishment. Not every disagreement is mutiny. Not every mistake proves malicious intent.

At the same time, "I was only joking" does not excuse firing weapons in occupied compartments, wasting vital supplies, opening hatches during flooding, tampering with the reactor, poisoning the crew, or interfering with navigation.

When frustration begins to affect your judgment, step away from the argument. Change compartments. Mute unnecessary communications. Request relief from a station when possible. A brief pause is preferable to a confrontation that ends in injury, arrest, or banishment at the next outpost.

The voyage is supposed to be survivable.

It may even, on rare occasions, be enjoyable.

DEPARTMENTAL RESPONSIBILITY

Every profession aboard a submarine exists because the vessel cannot survive without it.

Captain

The captain directs navigation, coordinates departments, selects missions, manages priorities, and accepts responsibility for the vessel's overall operation.

Engineer

Engineers maintain the reactor, electrical grid, junction boxes, batteries, and powered systems. They should be informed before anyone makes significant changes to the vessel's wiring or fuel configuration.

Mechanic

Mechanics repair the hull, pumps, engines, fabrication equipment, and other mechanical systems. They are generally among the first personnel required during flooding.

Medical Officer

Medical personnel preserve life, manage pharmaceuticals, diagnose injuries and infections, and establish treatment priorities during mass-casualty events. Medical storage is not a public dispensary.

Security Officer

Security protects the crew, controls access to weapons, responds to hostile boarding, investigates sabotage, and maintains order using the least force reasonably necessary.

Assistant

Assistants provide flexible labour wherever needed. A capable assistant may transport supplies, repair damage, support diving operations, reload weapons, recover casualties, and fill gaps between departments.

No profession exists merely to “fix things,” “shoot things,” or “annoy things.”

Learn the boundaries of your role, but do not use them as an excuse to ignore an obvious emergency. A security officer may fix a pump. A medic may carry ammunition. An assistant may weld a breach. An engineer may drag a casualty to medical.

Specialization keeps the submarine efficient.

Cooperation keeps it alive.

WHEN EVERYTHING GOES WRONG

There will be voyages during which several emergencies occur at once.

The hull breaches. The reactor begins overheating. Hostile creatures enter the ballast system. The medic is injured. The captain loses consciousness. Half the crew begins shouting contradictory reports over the radio.

When this happens, remember the order of survival:

1. Preserve breathable compartments.
2. Control flooding.

3. Stabilize the reactor and electrical systems.
4. Contain immediate hostile threats.
5. Recover and treat casualties.
6. Restore propulsion, navigation, and weapons.
7. Account for the crew.
8. Recover, reorganize, and resupply.

Do not run aimlessly between emergencies. Choose the most urgent problem you are capable of addressing and report that you are responding to it.

“Repairing aft breach.”

“Taking reactor.”

“Moving casualties to medical.”

“Reloading dorsal coilgun.”

Clear reports allow the captain to understand which problems remain unattended.

Panic spreads quickly through a submarine. So does competence.

Be the crew member whose arrival improves the situation.

This section fits after Leaving the Vessel and before the detailed medical and logistics chapters. It treats away missions as deliberate operations rather than spontaneous excursions.

AWAY MISSIONS

Mining, Salvage, Exploration, and Work Beyond the Hull

An away mission is any authorized operation requiring crew members to leave the submarine and perform a specific task in the European environment.

The purpose may be commercial, scientific, military, logistical, or merely necessary for the continued survival of the vessel. Away teams may be dispatched to mine mineral deposits, salvage wreckage, recover cargo, inspect ruins, investigate abandoned stations, collect biological specimens, repair external systems, eliminate hostile nests, or board another vessel.

Regardless of the objective, the same principle applies:

Leaving the submarine is not an ordinary change of workplace. It is entry into an environment permanently hostile to human life.

The submarine provides breathable air, pressure protection, shelter, electrical power, medical support, heavy weapons, navigation, and a defensible position.

The moment you pass through the airlock, most of those protections are reduced to whatever you are carrying on your person.

Prepare accordingly.

THE PURPOSE OF AN AWAY MISSION

Every departure from the submarine should have a clear objective.

That objective should be stated before the airlock opens.

Examples include:

- Mine a specific mineral formation.
- Recover cargo from a wreck.
- Search an abandoned vessel for survivors or supplies.
- Retrieve a mission item.
- Collect biological material.
- Inspect an alien ruin.
- Repair an external component.
- Clear a hostile nest.
- Establish whether a route is safe.
- Board and secure an enemy vessel.
- Recover a missing crew member or previous away team.

“Going outside to look around” is not a sufficiently defined mission unless the captain has expressly authorized reconnaissance.

Before departure, every member of the team should know:

- Where they are going.
- What they are expected to accomplish.
- Who is leading the operation.
- What equipment they are carrying.
- Which hazards are expected.
- How long the submarine intends to remain in position.
- What signal means the mission has been aborted.
- How the team will return.

An away mission does not require a lengthy ceremony. It does require enough planning that the crew does not discover, halfway through a flooded cave, that nobody brought a plasma cutter, ore bag, medical kit, or spare oxygen.

RESOURCE SPOTTING DURING TRANSIT

Mining is among the most common away-team operations conducted by independent submarines.

It is also one of the most important.

A vessel may survive through contracts alone for a time, but ammunition, medicine, fuel, replacement components, equipment, repairs, and crew wages steadily consume its income. Mineral recovery can mean the difference between a submarine that remains financially viable and one that spends every port call deciding which essential system it can no longer afford to support.

Crew members are therefore encouraged to remain observant during transit.

Anyone stationed at an external gun, camera, observation window, or periscope may notice mineral formations along cavern walls. These deposits may be valuable for direct sale or useful in fabricating ammunition, medicine, fuel, tools, electrical components, explosives, and other necessary supplies.

Reporting such a formation is optional unless resource spotting has been assigned as part of your duty.

You are not expected to abandon your post or continuously distract yourself from more important work merely to search for ore. During periods of calm, however, informing the engineering department or captain that a promising formation has been sighted may provide the vessel with a valuable opportunity.

Useful reports include:

- "Mineral formation on the lower cavern wall."
- "Possible uranium deposit beneath us."
- "Pyromorphite or similar ore visible near the aft quarter."
- "Large mixed deposit beside the cave entrance."
- "Resources visible, but the approach appears occupied."
- "Mineral field marked on sonar; recommend returning after the current mission."

Avoid presenting uncertain identification as fact.

From inside the submarine, poor lighting, distance, movement, and obstruction may make exact identification difficult. "Possible uranium" is more useful than confidently announcing uranium and sending the entire engineering department outside to discover several kilograms of something nearly worthless.

The captain must then decide whether stopping is worthwhile.

A deposit may be valuable, but the vessel may be under time pressure, carrying vulnerable cargo, running low on fuel, pursued by hostile forces, or navigating an area too dangerous for a prolonged stop.

The crew member who spots the resource provides information.

The captain decides whether the submarine can safely exploit it.

THE COST OF STOPPING

A mining opportunity must be considered against the danger of remaining stationary.

A submarine that stops beside a cavern wall becomes easier for hostile wildlife to approach. The engines no longer provide the same ability to evade contact. Diving personnel outside the hull limit the captain's freedom to manoeuvre. Weapon crews may have restricted firing angles because friendly divers are operating near the target area.

A rich mineral deposit can preserve the vessel's finances.

It can also keep the submarine in one place long enough to attract every unnatural abomination within several kilometres.

The away team should therefore work deliberately.

Do not spend ten minutes discussing which person should mine the first piece of ore. Do not wander into side passages while the submarine waits. Do not mine three fragments and then begin a recreational tour of the cave system.

Recover the intended resources, remain aware of the situation, and return when recalled.

The captain may terminate the operation at any time because of:

- Approaching hostile contacts.
- Changing current or terrain conditions.
- Reactor or electrical problems.
- Hull damage.
- Loss of communication.
- Mission time constraints.
- Low ammunition.
- Another crew emergency.
- The discovery of a larger threat nearby.

A recall order is not a suggestion.

The mineral deposit will still be present if the submarine survives long enough to return.

EQUIPPING THE AWAY TEAM

The equipment carried should match the mission.

Carrying too little leaves the team unable to complete its work. Carrying too much deprives the submarine of supplies, slows the diver, and concentrates a dangerous amount of valuable equipment on a person who may not return.

A basic away-team loadout may include:

- A functioning diving suit.
- A full oxygen tank.
- At least one reserve oxygen supply when the mission warrants it.
- An underwater scooter for longer journeys.
- A handheld sonar or navigation aid.
- A radio headset.
- A light source.
- The tools required for the objective.
- Basic medical treatment.
- A weapon appropriate to the expected threat.
- Storage for recovered materials.
- Spare batteries or fuel where necessary.

A mining team may require plasma cutters and ore storage.

A salvage team may require cutting tools, crowbars, wiring equipment, and additional carrying capacity.

A security operation may require weapons, ammunition, restraints, breaching tools, and medical support.

An expedition into an alien ruin may require all of the above, along with substantially more caution than the average crew member believes necessary.

Before cycling the airlock, check:

- Suit condition.
- Oxygen level.
- Radio function.
- Weapon condition.
- Ammunition.
- Tool batteries or fuel.
- Medical supplies.
- Storage capacity.
- Team readiness.

Do not discover that your oxygen tank is empty after leaving the submarine.

Do not take every spare oxygen tank aboard because you are afraid of running out.

Carry a reasonable reserve and leave enough equipment for the other people who may need to respond if your team encounters trouble.

TEAM ORGANIZATION

Away teams should avoid operating as an unstructured mass.

Even a small operation benefits from basic role assignment.

Depending upon the mission, the team may include:

Team Leader

The leader maintains communication with the submarine, directs movement, monitors the objective, and decides when the team should advance, withdraw, or request assistance.

Specialist

The specialist performs the principal task: mining, repairs, salvage, medical examination, scientific collection, or artifact recovery.

Security

Security watches the surrounding environment, protects the specialist, maintains a clear line of fire, and avoids becoming so fascinated by the work that an approaching crawler reaches the team unnoticed.

Medical Support

For dangerous or extended missions, a medic or a crew member carrying field treatment should accompany the group.

Porter or General Assistant

Recovered ore, cargo, equipment, and salvage must be carried. A general assistant can transport materials, manage oxygen reserves, help recover casualties, and support whichever task becomes urgent.

One person may perform several of these functions on a small team.

What matters is that someone is completing the objective while someone else is watching for the danger approaching from behind.

MAINTAINING A WATCH

Once outside the vessel, every crew member should remain aware of their surroundings.

Europa's oceans contain natural predators, territorial creatures, parasites, hostile human forces, pirates, separatists, rogue clowns, automated defences, unstable machinery, and entities whose classification is still disputed by several scientific institutions.

Threats may approach from above, below, behind, or from inside the structure the team is exploring.

Do not allow every member of the team to stare at the same mineral formation.

Do not let the entire group crowd around one storage container while nobody watches the corridor.

Do not become so absorbed in operating a plasma cutter that you fail to notice the creature biting the person beside you.

Maintain periodic visual checks.

Listen to the radio.

Watch the sonar when available.

Be alert for:

- Movement in dark water.
- Sudden changes in local wildlife.
- Damaged or opened doors.
- Fresh blood.
- Unexplained equipment.
- Disturbed silt.
- Electrical activity.
- Mechanical sounds.
- Acidic clouds.
- Unidentified growth.
- Missing team members.
- Radio silence.
- Objects that have moved since the last observation.

A person who sees a threat should report it immediately and clearly.

“Movement behind us.”

“Crawler above the entrance.”

“Possible human contact in the wreck.”

“Green cloud forming near the cave wall.”

“Unknown machine active inside the ruin.”

Do not scream incoherently into the radio and assume the team will understand what has happened.

Location matters.

Direction matters.

The nature of the threat matters.

BIOLOGICAL AND CHEMICAL HAZARDS

Not every danger in the European sea has teeth.

Some organisms defend themselves through toxic, acidic, infectious, or corrosive substances. Others create hazardous environments merely by existing.

Cave growths, ballast flora, parasitic organisms, and various native life-forms may release acidic or otherwise harmful material when damaged, disturbed, or placed under stress.

This material can injure exposed personnel and damage equipment.

Acid may degrade:

- Diving suits.
- Metallic tools.
- Plastic seals.
- Weapon components.
- Storage containers.
- Electrical equipment.
- Submarine hull structures.
- The crew member unfortunate enough to be standing in it.

One common warning sign is a cloudy green substance dispersing through the water.

Among professional medical and scientific personnel, this may be described using terms such as acidic discharge, corrosive particulate cloud, toxic biological effluent, or a similarly respectable expression.

Among working submariners, it is frequently called fart gas.

The medical and scientific communities generally dislike this name.

It remains useful because an experienced submariner hearing:

“The cave is full of fart gas,”

will usually understand that the area contains a visible, harmful cloud, that breathing or protective equipment must be checked, and that prolonged exposure may injure both personnel and equipment.

Informal terminology is acceptable when it communicates danger quickly.

It should not replace a useful report.

“Fart gas in the lower tunnel, green cloud expanding toward the exit” is significantly more valuable than merely shouting “gas” and disappearing from the radio.

When corrosive or toxic material is encountered:

- Avoid unnecessary contact.

- Do not remain inside the cloud longer than required.
- Monitor suit condition.
- Check oxygen and breathing equipment.
- Withdraw if equipment begins failing.
- Warn the submarine and other team members.
- Treat exposure promptly.
- Do not cut, burn, or attack the source without understanding how it may react.

Some biological growths become more dangerous when damaged.

The solution to an unpleasant organism is not always to shoot it repeatedly.

BALLAST FLORA AND RELATED GROWTHS

Ballast flora is particularly dangerous because it can infest the submarine itself.

An away team encountering similar growth outside the vessel should report it, especially if the organism may be carried back aboard on equipment, salvage, suits, or recovered materials.

Do not drag unidentified biological matter through the airlock without warning medical and engineering personnel.

Do not store it beside food, medicine, reactor supplies, or ballast controls.

Specimens may be valuable.

They may also attempt to colonize the vessel.

When returning with biological samples:

- Seal them in appropriate containers.
- Report what was collected.
- Identify where it was found.
- Keep the sample isolated until examined.
- Do not open it in a shared compartment.
- Do not consume it.
- Do not inject it.
- Do not introduce it into the ballast system to “see what happens.”

The submarine is not a laboratory unless the captain has specifically agreed that it is one.

Even then, it should remain a laboratory capable of reaching the next station.

MINING OPERATIONS

Mining appears simple.

Locate a deposit. Exit the submarine. Cut the material free. Carry it back.

The basic procedure is simple.

The environment is not.

Before mining begins, determine whether the submarine can hold position safely and whether its weapons have firing coverage that will not endanger the divers.

The mining team should approach the deposit together. Security should watch the surrounding water while the miners work.

Recover the material systematically.

Do not scatter ore across the cavern floor. Do not fill every personal inventory slot and then abandon mission tools because there is no room left. Use bags, crates, or other storage whenever available.

The team should communicate when:

- The deposit has been reached.
- Mining has begun.
- Hostile movement is observed.
- Oxygen is becoming low.
- Carrying capacity has been reached.
- The deposit is exhausted.
- The team is returning.

Mining teams should also understand the materials they recover.

Certain minerals may be valuable because they can be refined into reactor fuel, ammunition, medicines, explosives, or electrical components. Others may be worth more when sold directly.

The away team does not need to conduct a full market analysis at the cave wall.

It should avoid discarding potentially useful material because its value is not immediately obvious.

When carrying capacity becomes limited, consult the engineering department or captain regarding priorities.

WRECK AND SALVAGE OPERATIONS

Wrecks may contain cargo, tools, weapons, medicine, fuel, machinery, mission records, and survivors.

They may also contain flooding, electrical hazards, unstable reactors, armed occupants, parasites, structural collapse, hostile wildlife, and former crew members who are no longer entirely former.

Approach every wreck as an unknown environment.

Do not assume it is abandoned because no lights are visible.

Do not assume the occupants are dead because nobody answered the radio.

Do not assume the dead occupants will remain inactive.

Before entry:

- Inspect the exterior.
- Identify accessible hatches or breaches.
- Look for signs of movement.
- Check for weapon systems.
- Determine whether the wreck is stable.
- Establish a return route.
- Inform the submarine where the team is entering.

Inside the wreck, maintain formation and mark the route when necessary.

Avoid opening every door simultaneously. A sealed compartment may contain water pressure, fire, infection, hostile life, or a remarkably angry survivor.

Recover useful supplies, but remember the purpose of the mission.

An away team assigned to retrieve one cargo manifest should not keep the submarine waiting while it removes every chair, cup, and damaged coil from the wreck.

Prioritize:

1. Mission objectives.
2. Survivors.
3. Critical supplies.
4. Valuable or useful salvage.
5. Objects of uncertain value.
6. Decorative rubbish.

Do not allow greed to keep the team inside a deteriorating structure after conditions have become unsafe.

HOSTILE PERSONNEL AND BOARDING OPERATIONS

Away teams may encounter pirates, separatists, hostile station personnel, rogue mercenaries, or other armed humans.

Human enemies may be more dangerous than wildlife because they understand doors, cover, firearms, explosives, deception, and surrender.

Maintain clear lines of fire.

Do not fire through teammates.

Do not allow the team to bunch tightly in a doorway.

Do not throw explosives into a compartment until the position of every friendly crew member is known.

During an assault, communicate movement:

- "Taking the left side."
- "Crossing."
- "Holding fire."
- "One hostile behind the door."
- "Friendly ahead."
- "Room secure."
- "Casualty."
- "Withdrawing."

A boarding operation is not improved by every person charging through the same hatch while firing over one another's shoulders.

Security should control the pace of entry. Medical personnel should remain protected but accessible. Specialists should not advance beyond security merely because they are eager to inspect a console or container.

If the hostile force withdraws, do not pursue blindly into an unknown structure.

They may know the terrain.

You do not.

ALIEN RUINS

Alien ruins are among the most valuable and dangerous locations an away team may encounter.

Their origin remains uncertain. Their purpose is often unclear. Their internal systems may have remained active for an unknown length of time beneath the European ice.

Crews have entered such ruins seeking artifacts, scientific data, rare materials, or station contracts.

Some have returned wealthy.

Some have returned altered.

Some have not returned.

Entire submarines have gone dark after disturbing mechanical entities released from alien structures. Reports describe automated guardians, energy discharges, shifting internal systems, hostile machines, unusual biological manifestations, and devices that appear inactive until a crew member approaches.

Treat every ruin as an active installation until proven otherwise.

Do not assume age means safety.

Do not assume silence means inactivity.

Do not assume a closed structure is unoccupied.

When entering:

- Maintain communication with the submarine.
- Keep track of the exit route.
- Avoid splitting the team unnecessarily.
- Watch for mechanical movement.
- Do not touch every visible device.
- Do not activate controls merely to observe the result.
- Maintain sufficient oxygen for the return journey.
- Withdraw if unfamiliar systems begin powering up.
- Report any defensive construct immediately.

Ruins should be explored slowly enough to observe danger but quickly enough that the submarine is not left exposed indefinitely.

Curiosity is part of the mission.

Survival remains the priority.

ARTIFACT HANDLING

Alien artifacts are frequently worth substantial amounts of money.

Scientific teams, research institutions, military offices, and private collectors may pay exorbitant prices for functioning examples of alien technology.

There is a reason for this.

Artifacts may produce heat, radiation, electrical interference, pressure anomalies, structural damage, psychological effects, biological reactions, or other consequences that remain poorly understood.

Some artifacts appear stable until moved.

Others react to water, air, electricity, nearby machinery, living organisms, or other artifacts.

A device operated in the wrong environment may produce an extremely exothermic response—which is the scientific way of saying that it may become catastrophically hot, ignite its surroundings, damage the hull, cook the carrier inside their suit, or cause the submarine to experience a brief and educational disaster.

Do not activate alien technology casually.

Do not strike it with tools.

Do not connect it to the submarine's electrical grid.

Do not carry several unknown artifacts together merely because they fit inside the same crate.

Do not place one beside the reactor to determine whether it “likes power.”

Before recovering an artifact:

- Observe it from a distance.
- Check for visible activity.
- Determine whether it is secured to the structure.
- Confirm the team has a safe carrying method.
- Warn the submarine that an artifact is being recovered.
- Keep it isolated from critical systems.
- Monitor the carrier for unusual effects.
- Be prepared to abandon it if it threatens the team or vessel.

No artifact is worth the submarine.

No salvage payment is high enough to compensate a crew that never reaches the station.

When an artifact begins producing heat, interference, radiation, electrical discharge, inexplicable sound, movement, or effects upon nearby life, report it immediately.

Do not conceal the problem because you are afraid the captain will order you to leave the valuable object behind.

The captain probably will.

That may be why the crew survives.

COMMUNICATION OUTSIDE THE SUBMARINE

The submarine must know where the away team is and what condition it is in.

Away personnel should report at meaningful stages of the operation:

- Departure from the airlock.
- Arrival at the objective.
- Discovery of a major hazard.
- Contact with hostile forces.
- Discovery of survivors.
- Completion of the objective.
- Beginning the return.
- Entry into the airlock.
- Final accounting of personnel.

Do not provide constant commentary about every stone, fish, door, and empty container.

Do not remain silent for the entire mission.

The purpose of communication is to keep the team coordinated and allow the submarine to respond if conditions change.

Useful reports are brief and specific.

“Mining team at the deposit. No contacts.”

“Wreck entered through upper breach.”

“Hostile life in the western tunnel.”

“One diver injured; returning.”

“Artifact recovered. It is emitting heat.”

“Team inside the airlock. All personnel accounted for.”

If communication fails, the away team should follow the contingency established before departure.

This may mean returning immediately, waiting at a known location, or using another signal.

Do not continue deeper into an unknown ruin after losing all contact unless the mission and emergency circumstances clearly justify it.

OXYGEN, DISTANCE, AND THE RETURN JOURNEY

A reserve oxygen tank is not additional exploration time.

It is insurance against delay.

Plan the mission so that the team can return before reaching its final supply. Movement may become slower because of casualties, heavy salvage, damaged equipment, hostile pursuit, difficult terrain, or a lost route.

A diver who begins returning only after the current tank is nearly empty has not preserved a reserve.

They have merely postponed suffocation.

Monitor oxygen regularly.

Do not rely upon another crew member to remind you.

If someone reports low oxygen, acknowledge the problem and begin planning the return. Do not tell them to endure a little longer while the team searches one final cabinet.

A recovered artifact is not valuable enough to justify carrying it while abandoning the person who helped retrieve it.

When necessary, leave salvage behind and recover it later.

The living crew can conduct another expedition.

The dead generally cannot.

CASUALTY RECOVERY

If a crew member is injured outside the submarine, the team must decide whether treatment can be safely provided at the location or whether the casualty must be evacuated.

Control immediate bleeding. Restore oxygen. Remove the casualty from continuing danger. Use field supplies to stabilize them.

Do not administer every drug in the medical kit merely because the team is frightened.

Report:

- Who is injured.
- The nature of the injury.
- Whether the casualty is conscious.
- Whether the area remains dangerous.
- Whether the team can return without assistance.

Examples:

- “Miner injured by crawler. Moderate bleeding controlled. Returning.”
- “Diver unconscious with oxygen loss. Medical needed at airlock.”
- “Security has a GSW. Hostiles still active.”
- “One casualty immobile. Request recovery team.”

The submarine should prepare medical assistance before the casualty arrives.

Do not drag an injured diver into the airlock and only then inform the medical officer that they have not been breathing for several minutes.

If a casualty cannot be recovered without causing additional deaths, the team leader may order withdrawal.

This is a grave decision.

It is sometimes necessary.

Europa readily turns one casualty into an entire missing team when rescuers abandon judgment.

ABORTING THE MISSION

An away mission may be aborted for any reasonable threat to the team or submarine.

Reasons include:

- Insufficient oxygen.
- Serious injury.
- Loss of communication.
- Overwhelming hostile contact.
- Equipment failure.
- Unexpected radiation or chemical exposure.
- Artifact instability.
- Activation of alien defences.
- The submarine coming under attack.
- Loss of the route back.
- Direct recall from the captain.

Aborting a mission is not cowardice.

It is recognition that the objective cannot presently be completed at an acceptable cost.

The team may return later with additional equipment, personnel, ammunition, knowledge, or a better-positioned submarine.

A dead crew cannot make a second attempt.

When an abort is ordered:

1. Acknowledge it.
2. Gather personnel.
3. Recover essential equipment if immediately possible.
4. Leave nonessential salvage.
5. Follow the safest known route.
6. Report progress toward the submarine.
7. Account for everyone before departure.

Do not delay withdrawal because one crew member insists upon retrieving a final mineral fragment.

Do not conceal an abort from the submarine while attempting to complete the mission anyway.

RETURNING ABOARD

The away mission is not complete when the team reaches the airlock.

Before the submarine resumes movement, confirm:

- Every living team member has returned.
- Casualties have been delivered to medical.
- Recovered equipment has been secured.
- Dangerous specimens have been isolated.
- Artifacts have been reported.
- Weapons have been made safe.
- Empty oxygen tanks have been separated for refilling.
- Mission-critical items have been delivered.
- No one remains outside without authorization.

Report the team's return clearly.

"All divers aboard."

"One casualty in medical."

"Artifact secured in the forward storage compartment."

"Mining team returned with six containers."

“Two members aboard; third team member still outside.”

Do not answer “everyone is aboard” merely because you personally have returned.

The captain needs an accurate accounting before moving the submarine.

Once the team is safe, conduct a brief inventory. Report what was recovered, expended, lost, damaged, or discovered.

This information will affect the next port call and the next away mission.

FINAL RULE OF THE AWAY TEAM

Away missions keep submarines alive.

They provide minerals, medicine, salvage, artifacts, intelligence, cargo, and access to places the vessel itself cannot enter. They are also among the most dangerous operations a crew can undertake.

A successful away-team member is not the person who travels farthest into the cave, carries the strangest artifact, kills the most creatures, or returns with the most dramatic story.

It is the person who helps complete the objective and brings the team home.

Watch the water.

Watch your oxygen.

Watch your line of fire.

Listen to the recall.

Do not touch what you do not understand.

Do not remain for one more piece of salvage when the situation has already become unsafe.

Beyond the submarine’s hull, Europa does not distinguish between bravery, greed, curiosity, and stupidity.

It consumes all four with equal efficiency.

The revised version keeps the captain’s chapter aimed at experienced operators, but gives it more of the dry, fatalistic humor expected from people who voluntarily command nuclear pressure vessels through monster-infested caves.

THE OXYGEN GENERATOR

The Loudest Thing Nobody Remembers to Turn Off

When captains discuss the acoustic profile of a submarine, attention generally falls upon the obvious offenders.

The reactor hums.

The engines thunder.

The pumps hammer.

The guns announce themselves with all the subtlety of a collapsing station.

Yet repeated acoustic surveys have identified another machine as one of the loudest, most distinctive, and most persistent sources of detectable noise aboard the average European submarine:

The oxygen generator.

This tends to surprise new captains.

It should not surprise experienced ones, although it frequently does.

The oxygen generator is active almost continuously. It draws substantial electrical power, cycles pumps and processing machinery, moves atmosphere through the vessel, separates useful gases, and performs the ongoing technological miracle required to keep an entire crew from quietly becoming corpses.

It is less a household appliance and more an industrial atmospheric plant compressed into a compartment barely large enough for an engineer to swear inside.

Naturally, it is loud.

THE SUBMARINE'S ACOUSTIC BEACON

A functioning oxygen generator produces a persistent mechanical and electrical signature.

This signature may include:

- Compressor vibration.
- Fluid circulation.
- Ventilation noise.

- Electrical switching.
- Pump resonance.
- Atmospheric-processing machinery.
- Hull-borne vibration.
- Low-frequency electrical emissions.
- Extremely low-frequency harmonic interference.
- The characteristic rattling noise engineering insists is “within tolerance.”

Together, these sounds produce an identifiable acoustic pattern capable of travelling considerable distances through the European sea.

The exact mechanism remains the subject of prolonged technical disagreement.

Some researchers attribute the effect primarily to conventional vibration passing through the hull. Others identify unusual ELF and EXF emissions generated by the high-energy electrical systems involved in oxygen production, gas separation, atmospheric balancing, and thermal regulation.

Several papers insist that EXF is not a recognized classification.

Several captains insist the creatures do not care what the papers call it.

Whatever the theoretical explanation, practical observation remains consistent:

A powered oxygen generator makes the submarine easier to detect.

It may be quieter than an engine operating at full thrust or a railgun firing nuclear ammunition.

It is not quieter than the same submarine with the generator switched off.

WHY OXYGEN PRODUCTION IS SO NOTICEABLE

The machinery used aboard modern submarines does more than simply pull breathable oxygen out of nowhere.

Depending upon the submarine’s design, atmospheric generation may involve:

- Electrolysis of processed water.
- Oxygen separation and concentration.
- Carbon-dioxide scrubbing.
- Removal or balancing of inert gases.
- Atmospheric circulation.
- Thermal control.
- Pressure regulation.
- Waste-gas venting.

- Continuous chemical and electrical monitoring.

Water electrolysis separates hydrogen and oxygen. Additional atmospheric systems then regulate the vessel's gas mixture, manage contaminants, and maintain conditions the crew can breathe without becoming unconscious, combustible, or scientifically interesting.

This process requires energy.

Considerable energy.

The generator is effectively conducting a controlled industrial reaction powerful enough to maintain breathable conditions for an entire submarine full of moving, panicking, metabolically active personnel.

That process creates heat, vibration, electromagnetic interference, and acoustic resonance.

Who could have imagined that an ongoing electrical argument violent enough to manufacture the atmosphere for an entire crew might be audible outside the vessel?

Apparently not the mop heads who designed the first generation without adequate acoustic baffling.

SILENT RUNNING

When avoiding sound-sensitive predators or hostile detection systems, the captain may order the oxygen generator depowered.

This can significantly reduce the submarine's acoustic footprint.

It can also place every living person aboard on a timer.

The moment oxygen production stops, the submarine begins consuming the breathable atmosphere already present inside its compartments. The precise safe duration depends upon:

- Crew size.
- Submarine volume.
- Existing oxygen level.
- Number of flooded or sealed compartments.
- Condition of ventilation.
- Fires.
- Hull breaches.
- Crew exertion.
- Use of diving masks or suits.
- Whether someone has decided silent running is the ideal time for a sprinting competition.

A large, lightly crewed vessel may remain habitable for a considerable period.

A small submarine packed with personnel may consume its breathable reserve much faster.

The captain must understand that disabling oxygen generation is not free stealth.

It is the exchange of one danger for another.

THE NO-OXYGEN ALARM

Most submarines are fitted with an alarm that activates when atmospheric oxygen falls below safe levels or when oxygen generation is disabled.

Some captains object that the alarm defeats the purpose of silent running.

Acoustic testing indicates otherwise.

The alarm may be irritating.

It may be repetitive.

It may provoke the crew to report the obvious every twelve seconds.

It is still generally quieter, in terms detectable through the hull and surrounding water, than the oxygen-generation machinery it warns has been disabled.

This is because the alarm is intended for human ears inside the submarine.

The oxygen generator transmits vibration through the vessel's structure, electrical systems, pipework, and surrounding water. To an organism sensitive to mechanical or electrical activity, the generator may resemble a continuous beacon.

The alarm is a bell ringing inside a box.

The oxygen generator is the box itself humming across the ocean.

Do not reactivate the generator solely because the crew complains that the alarm is annoying.

Do reactivate it before the crew becomes too hypoxic to complain coherently.

ACOUSTIC NOISE AND SONAR DETECTION

The oxygen generator contributes to two related but distinct detection problems.

First, it raises the submarine's general acoustic signature. Native predators may follow vibration, electrical disturbance, heat, or machinery noise even when active sonar is disabled.

Second, the generator may contribute to what the submarine's own detection systems perceive as background noise. Depending upon the vessel's layout, maintenance condition, and sensor isolation, internal machinery can make faint external contacts more difficult to distinguish.

A captain attempting to run silently while leaving every internal system operating at full load is practicing a very selective interpretation of silence.

For reduced-signature operation, consider the condition of:

- Oxygen generation.
- Engines.
- Pumps.
- Fabricators.
- Deconstructors.
- Battery chargers.
- Weapon capacitors.
- Active sonar.
- Unnecessary machinery.
- Alarms and ventilation.
- The assistant repeatedly striking the hull with a wrench to "see if anything answers."

The oxygen generator is rarely the only detectable system.

It is simply one of the most frequently overlooked.

WHEN TO DEPOWER THE GENERATOR

Disabling oxygen generation may be appropriate when:

- A large sound-sensitive creature is nearby.
- The submarine is hiding from hostile vessels.
- Passive sonar contact suggests pursuit.
- The captain intends to drift silently past a threat.
- Active sonar has been disabled for concealment.
- The submarine is holding position during an ambush.
- A mission specifically requires minimal acoustic output.
- Engineering identifies the generator as worsening an electrical or thermal emergency.

The captain should inform the crew before shutting it down.

Useful announcements include:

- "Silent running. Oxygen generator off."
- "Conserve air and limit exertion."
- "Wear masks if assigned."
- "Report low-oxygen symptoms immediately."

- “Engineering, monitor atmospheric levels.”
- “Generator remains offline until we clear the contact.”

Do not switch off the oxygen generator without explanation and then act surprised when engineering assumes it has broken.

WHEN NOT TO DEPOWER IT

Silent running is not appropriate merely because a contact exists somewhere on sonar.

Do not disable oxygen production when:

- The crew is already suffering low oxygen.
- Fire is consuming the atmosphere.
- Large portions of the vessel are flooded.
- Several compartments are isolated from ventilation.
- Casualties require reliable breathable air.
- The crew lacks sufficient masks or oxygen tanks.
- The threat has already detected and engaged the submarine.
- The submarine must use maximum engine power immediately.
- No one is monitoring atmospheric conditions.
- The captain has forgotten why the generator was switched off.

Once the submarine is actively fighting, flooding, burning, and firing every weapon, the acoustic benefit of keeping the generator offline may become negligible.

At that point, restoring breathable air is generally more useful than preserving the delicate secrecy already destroyed by the railgun.

THE CAPTAIN’S OXYGEN CLOCK

When the generator is depowered, the captain should begin treating the atmosphere as a finite operational resource.

Monitor oxygen levels.

Ask engineering for updates.

Account for crew activity.

Set a decision point for reactivation.

Do not wait until several crew members collapse before remembering that breathing was included in the voyage plan.

A useful sequence is:

1. Announce silent running.
2. Disable active sonar and unnecessary machinery.
3. Depower the oxygen generator.
4. Reduce movement and crew exertion.
5. Monitor atmospheric levels.
6. Track the hostile contact.
7. Determine when the submarine has cleared the danger.
8. Restore oxygen generation.
9. Confirm oxygen levels are recovering.
10. Inform medical if anyone became symptomatic.

Silence without a reactivation plan is merely suffocation conducted professionally.

CREW CONDUCT DURING OXYGEN CONSERVATION

When oxygen generation is offline, the crew should reduce unnecessary exertion.

This means:

- Do not sprint through the submarine without need.
- Do not engage in recreational fights.
- Do not start unnecessary fabrication.
- Do not begin nonessential welding.
- Do not crowd into a small sealed compartment.
- Do not remove masks from emergency storage for personal decoration.
- Do not light fires.
- Do not respond to the oxygen alarm by shouting continuously into the radio.

Crew members should report symptoms such as:

- Dizziness.
- Confusion.
- Weakness.
- Shortness of breath.
- Loss of coordination.
- Unusual fatigue.
- Impaired judgment.
- Sudden confidence in a plan involving the airlock.

Medical personnel should remember that hypoxia can affect behavior before a crew member loses consciousness.

The person insisting they feel perfectly fine while attempting to repair the reactor with a diving knife may not, in fact, be perfectly fine.

ENGINEERING RESPONSIBILITIES

Engineering should maintain the oxygen generator in good repair and understand how its operation affects both power demand and acoustic output.

Responsibilities include:

- Repairing generator damage.
- Monitoring its electrical load.
- Checking atmospheric circulation.
- Maintaining pumps and associated machinery.
- Confirming ventilation into occupied compartments.
- Reporting abnormal noise or vibration.
- Assisting the captain with atmospheric estimates.
- Restoring operation promptly after silent running.
- Determining whether the generator is louder than normal because something has begun coming apart inside it.

An oxygen generator becoming unusually loud may indicate mechanical wear, damaged components, poor mounting, failing pumps, or electrical instability.

Do not dismiss every new grinding noise as part of its charming personality.

A badly maintained generator may become louder, less efficient, and more likely to fail precisely when the crew needs it most.

MEDICAL RESPONSIBILITIES

Medical personnel should monitor the crew during prolonged oxygen shutdowns.

Priority should be given to:

- Existing casualties.
- Crew members with blood loss.
- Unconscious personnel.
- Those exposed to smoke.
- Divers returning with depleted oxygen.

- Personnel performing heavy physical work.
- Anyone already experiencing respiratory distress.

A crew member with severe blood loss may tolerate reduced oxygen poorly because the blood remaining in their body is already less capable of supplying their tissues.

Restoring breathable atmosphere does not automatically correct every effect of prolonged hypoxia.

Medical should remain prepared to provide oxygen and treatment after the generator resumes operation.

THE LIMITS OF SILENCE

Turning off the oxygen generator does not make the submarine invisible.

The vessel may still be detected through:

- Engine noise.
- Pump operation.
- Reactor activity.
- Electrical systems.
- Active sonar.
- Weapon discharge.
- Hull vibration.
- Heat.
- Movement.
- External lights.
- Crew activity.
- The enormous creature already chewing on the engine compartment.

Silent running reduces detection.

It does not grant immunity.

The captain must still navigate carefully, monitor contacts, and decide whether concealment remains useful.

Do not remain oxygen-starved for ten minutes after the threat has clearly begun attacking.

At that point, secrecy has concluded.

FINAL RULE OF OXYGEN SILENCE

The oxygen generator is one of the submarine's most vital machines.

It is also one of its loudest continuous advertisements.

Depowering it can help a vessel disappear into Europa's black water.

It can also turn the submarine into a sealed container full of increasingly confused people.

Use oxygen silence deliberately.

Inform the crew.

Monitor the atmosphere.

Know the threat.

Know the time.

Know when to turn the machine back on.

The purpose of silent running is to avoid being eaten.

Achieving perfect acoustic concealment only to suffocate the entire crew is technically successful silence, but captains are advised not to include it in the mission report.

THE CAPTAIN'S CHARGE

Command, Navigation, Crew Welfare, and the Ancient Maritime Art of Pretending Everything Is Under Control

The captain's first duty is to keep the crew alive.

The captain's second duty is to maintain forward momentum.

The captain's third duty is to convince everyone aboard that the first two duties remain achievable, even when the reactor is complaining, engineering is waist-deep in water, security has misplaced a shotgun, medical has begun asking whether anyone knows where the blood went, and the sonar display contains a contact large enough to qualify for its own postal district.

A captain is not merely the person sitting at the helm.

They are navigator, coordinator, employer, quartermaster, negotiator, disciplinarian, mission planner, emergency dispatcher, and—during particularly difficult voyages—the only person aboard still pretending to have read the instruction manual.

The captain does not personally repair every junction box, treat every casualty, reload every weapon, or count every fuel rod.

They are responsible for knowing who is supposed to be doing those things, whether those people are still alive, and why none of them have answered the radio.

Command is not the art of doing everything yourself.

It is the art of ensuring everything necessary is being done by someone, somewhere, preferably before it catches fire.

I. THE FUNDAMENTAL RESPONSIBILITY OF COMMAND

A captain must balance survival against progress.

A submarine that never leaves port is safe, but it earns nothing and eventually becomes an unusually expensive apartment.

A submarine driven from mission to mission without pause may earn quickly, but it will eventually exhaust its fuel, ammunition, medicine, engineering supplies, crew loyalty, or all five at once.

The captain must maintain momentum without spending the vessel's future to accelerate its present.

At any given moment, command should have a reasonable understanding of:

- The current mission.
- The next mission.
- The route between them.
- The condition of the hull.
- The state of the reactor.
- Available fuel.
- Remaining ammunition.
- Medical reserves.
- Engineering materials.
- Oxygen and welding fuel.
- The condition of pumps, engines, and ballast systems.
- The position of hostile contacts.
- The location of away teams.
- The location of the crew members who were specifically told not to join the away team.
- The amount of operating time remaining before something essential runs out.

The captain must think beyond the next few minutes.

A submarine may presently be intact while possessing insufficient fuel to reach the next station. It may have ammunition but no medicine, medicine but no ammunition, or plenty of both but no mechanic capable of keeping the sea on the correct side of the hull.

A good captain asks:

“Can we continue?”

A better captain also asks:

“Can we continue and still have enough fuel to regret the decision later?”

II. KNOW YOUR DEPARTMENT HEADS

Every major function aboard the submarine should have a responsible officer or senior hand.

Depending upon the size of the vessel, this may include:

- Chief engineer.
- Senior mechanic.
- Medical officer.
- Security chief.
- Gunnery officer.
- Quartermaster or logistics officer.
- Away-team leader.
- Executive officer or relief navigator.

On a small submarine, one person may hold several of these responsibilities.

This is efficient until that person is eaten.

The captain should know:

- Who is operating the reactor.
- Who is repairing the hull.
- Who is responsible for medical.
- Who is coordinating security.
- Who is manning the guns.
- Who is outside.
- Who is unconscious.
- Who is detained.
- Who is missing.
- Who has command if the captain is rendered unable to continue.

Undefined responsibility produces a familiar type of emergency in which four people repair the same pump while nobody notices the reactor compartment filling with water.

Delegate clearly.

Do not say:

“Someone should probably deal with that.”

Say:

- “Chief engineer, stabilize the reactor.”
- “Mechanics, repair aft ballast.”
- “Security, protect engineering.”
- “Medical, prepare for casualties.”
- “Gunnery, cover the lower approach.”
- “Away team, return immediately.”
- “Assistant, stop carrying that chair and bring welding fuel.”

The difference between a suggestion and an order is often several tonnes of water.

III. KEEP A ROSTER OR BEGIN COLLECTING GHOSTS

The captain must know who is aboard.

This sounds simple until departure, when someone asks where the medical officer is and receives six different answers:

- “Medical.”
- “The station bar.”
- “The wreck.”
- “Unconscious.”
- “I thought you had them.”
- “There was a medical officer?”

Crew members may be:

- At their stations.
- Moving between compartments.
- Outside the submarine.
- Aboard a shuttle.
- Inside a wreck or ruin.
- On station leave.
- Injured.
- Detained.
- Dead.
- Missing under circumstances that will later become a disciplinary hearing.

Before leaving a station, wreck, cave, ruin, or mission site, account for the crew.

Use names, departments, or team reports.

Do not accept “everyone is probably aboard” as a final accounting.

Probability is useful in mathematics.

It is less useful when the missing person is carrying the only plasma cutter.

IV. KNOW WHO IS OUTSIDE THE HULL

The moment a crew member leaves the submarine, the captain's freedom to manoeuvre becomes limited.

Divers may be mining beside the hull, exploring caves, attached to wreckage, crossing weapon arcs, or attempting to recover something described only as "very valuable" and "probably safe."

The captain should know:

- How many people are outside.
- Who leads them.
- Their objective.
- Their expected duration.
- Their approximate position.
- Whether they are near external weapon arcs.
- Whether the submarine can move without dragging them into terrain.
- Whether hostile contacts are approaching.

Before moving, warn the away team.

Before firing heavy weapons, locate the away team.

Before departing, count the away team.

A captain who forgets divers does not merely lose personnel.

They establish a reputation.

Reputation travels quickly between stations, particularly when the story begins with:

"They left four miners behind because the captain thought the green lights meant everyone was aboard."

V. FORWARD MOMENTUM

The captain is responsible for keeping the submarine moving toward useful objectives.

This does not mean keeping the engines at full power simply so the scenery changes faster.

Forward momentum includes:

- Completing contracts.
- Reaching new stations.
- Improving the submarine.
- Maintaining financial solvency.
- Recovering resources.
- Building crew experience.
- Expanding into more distant waters.
- Avoiding pointless engagements.
- Withdrawing from missions that have become ruinously expensive.
- Preparing for longer routes.

Sometimes progress means advancing.

Sometimes it means stopping to mine uranium.

Sometimes it means remaining in port while engineering explains, for the third time, that one fuel rod is not a strategic reserve.

Sometimes it means retreating before a difficult mission becomes a permanent landmark on someone else's sonar.

Motion is not always progress.

A submarine falling toward crush depth is moving very efficiently.

VI. NAVIGATION

The captain's most visible responsibility is guiding the vessel from point A to point B without discovering an uncharted point C inside a cavern wall.

Navigation requires awareness of:

- Cavern ceilings.
- The seabed.
- Narrow passages.
- Ice formations.
- Wreckage.
- Ruins.
- Hostile contacts.
- Mineral deposits.
- Mission objectives.
- Docking structures.
- Current depth.
- Ascent and descent rates.
- Ballast authority.
- Engine response.
- Hull protrusions.

- Any poorly placed exterior gun whose barrel extends farther than the designer admitted.

The captain must navigate the entire submarine, not merely the centre marker displayed on the helm.

The central point may pass cleanly through a gap while the engine, railgun, docking port, or decorative observation blister removes itself against the rock.

Submarine designers refer to this as “operational clearance.”

Captains refer to it as “that bloody corner.”

VII. THE LEGEND OF THE BLIND CAPTAIN

Veteran captains occasionally swear upon their epaulets that they can navigate Europa without active sonar.

They claim to follow cavern roofs by engine echo, hull vibration, water movement, intuition, or an ancestral maritime sense inherited from relatives who once operated fishing boats on Earth.

Some of them can.

Some of them merely collide with confidence.

Running without active sonar may be appropriate when avoiding creatures sensitive to sound or attempting to conceal the vessel's position. It should not become a theatrical demonstration of seniority.

Navigation information may come from:

- Active sonar.
- Passive sonar.
- Cameras.
- Periscopes.
- Viewports.
- Gunnery reports.
- Away-team observations.
- Known charts.
- Prior experience.
- The mechanic shouting that the wall is getting closer.

When operating without active sonar:

- Reduce speed.
- Assign visual observers.
- Use passive contacts.
- Follow known terrain.
- Request reports from gunnery.
- Use brief sonar pulses when necessary.

- Resist the urge to prove that rocks respect rank.

Navigating by instinct is impressive.

Navigating by instinct while ignoring a fully functional sonar set is how captains become cautionary anecdotes.

VIII. SONAR DISCIPLINE

Active sonar allows the submarine to see.

It also allows much of Europa to see the submarine.

The captain must decide whether the immediate danger is blindness or attention.

Available options include:

- Continuous active sonar.
- Periodic pulses.
- Directional sonar.
- Passive listening.
- External observation.
- Reduced-speed silent running.
- Asking the gun crews whether anything enormous appears to be following the vessel.

The crew should know when active sonar is disabled.

Useful commands include:

- "Active sonar off. Gunnery maintain visual watch."
- "One short pulse."
- "Large contact below."
- "Running silent until clear."
- "Resume active sonar."
- "Someone explain why the contact is now above us."

The captain need not narrate every sonar return.

They should report anything likely to alter the crew's remaining life expectancy.

IX. HAZARD AVOIDANCE

Not every creature must be fought.

Not every cave must be entered.

Not every ruin must be investigated.

Not every mineral formation justifies parking the submarine beside what appears to be a nest containing every crawler born since settlement began.

The captain should weigh:

- Threat size.
- Threat speed.
- Ammunition reserves.
- Weapon coverage.
- Hull condition.
- Crew readiness.
- Nearby terrain.
- Escape routes.
- The presence of divers.
- Mission urgency.
- Potential reward.
- Whether the thing on sonar appears to have noticed the submarine.

Avoidance is not cowardice.

It is ammunition conservation performed at engine speed.

A captain who hunts every contact will eventually find one that has also been hunting.

X. SPEED AND THE SCIENCE OF ARRIVING IN ONE PIECE

Maximum engine output is not always the fastest route to success.

High speed reduces reaction time, increases collision risk, strains the reactor, and makes docking resemble an insurance fraud attempt.

High speed may be appropriate:

- In open water.
- During escape.
- During a timed mission.
- On a known route.
- When something behind the vessel has more teeth than the vessel has ammunition.

Reduced speed is appropriate:

- Near terrain.
- Near stations.

- Near wrecks.
- With divers outside.
- During silent running.
- When ballast control is impaired.
- During reactor instability.
- When operating an unfamiliar submarine.
- Whenever the mechanic says, "Please stop doing that."

The engine control is not a moral test.

A captain may reduce speed without surrendering their epaulets.

XI. REACTOR AND FUEL AWARENESS

The captain need not personally operate the reactor.

They should understand what it means when engineering says it is "fine."

"Fine" may mean:

- Stable.
- Temporarily stable.
- On automatic control.
- Being manually held together.
- One load spike away from becoming educational.
- The engineer does not currently have time to explain.

The captain should know:

- Who is operating the reactor.
- Whether it is on manual or automatic control.
- The condition of the active fuel rod.
- The number of reserves.
- Whether the grid is stable.
- Whether heavy systems are drawing excessive power.
- How long the vessel can continue.
- Whether more fuel can be fabricated.

Ask for useful reports.

"One rod active at forty percent, two reserves, and enough uranium for another."

This is useful.

"Probably enough."

This is not a fuel report.

It is a prophecy.

Fuel planning must include the return journey, delays, combat, detours, and the possibility that the next station has no rods for sale because another captain purchased all of them after receiving the same prophecy.

XII. SUPPLIES AND THE CAPTAIN'S BROADLY INFORMED ANXIETY

The captain should maintain a general understanding of the vessel's stores.

They need not personally count every bandage.

They should know whom to ask and whether that person has recently been using the bandages as decorative streamers.

Important supplies include:

- Reactor fuel.
- Welding fuel.
- Oxygen.
- Ammunition.
- Medical treatment.
- Blood products.
- Mechanical components.
- Electrical components.
- Fabrication materials.
- Diving equipment.
- Batteries.
- Tools.
- Anything the crew insists is "mission critical" after purchasing it personally.

The captain should also know what can be fabricated aboard.

Minerals may become ammunition, medicine, fuel, explosives, tools, or electrical components.

They do not become those things through optimism.

The vessel still requires working machinery, power, trained personnel, and time.

A cargo hold full of minerals is not the same as a stocked submarine.

It is a stocked submarine after engineering has spent twenty minutes arguing about fabrication priorities.

XIII. MISSION SELECTION

The highest-paying mission is not always the most profitable.

A generous contract may be offset by:

- Ammunition expenditure.
- Hull repairs.
- Medical treatment.
- Fuel consumption.
- Crew replacement.
- Equipment loss.
- Reputation damage.
- Fines.
- Bribes.
- An artifact deciding that the engine room should become hotter.

Mission selection should account for:

- Crew skill.
- Vessel capability.
- Distance.
- Supplies.
- Threats.
- Political consequences.
- Salvage opportunities.
- Mining opportunities.
- Whether objectives can be combined.
- Whether the destination is likely to purchase the things the crew will inevitably steal along the way.

A captain who accepts every contract may appear ambitious.

They may also discover that escorting a transport, killing a swarm, recovering an artifact, clearing a nest, and delivering volatile cargo do not become easier when attempted simultaneously.

Take work the submarine can complete.

There is no shame in declining a mission.

There is considerable shame in accepting one, losing half the crew, and discovering the payment does not cover the ammunition.

XIV. COMMUNICATION FROM THE HELM

The captain should provide enough information for the crew to understand what the submarine is doing.

The crew does not require a poetic narration of every adjustment.

It should know when:

- Danger is approaching.
- Speed is changing.
- Depth is changing.
- Sonar is being disabled.
- Heavy weapons may be needed.
- Engineering should expect increased load.
- Divers must return.
- Docking has begun.
- The vessel is retreating.
- Mission priorities have changed.
- Someone should check why medical is on fire.

Useful commands are concise:

- "Large contact below. Gunnery prepare."
- "Engineering, maximum engine load incoming."
- "Away team recall."
- "Mechanics to aft breach."
- "Running silent."
- "Medical, prepare for casualties."
- "All departments report readiness."

The captain should also listen.

Command does not improve merely because the captain has successfully occupied every radio frequency at once.

XV. MORALE AND THE MAINTENANCE OF A CREW THAT DOES NOT HATE YOU

A submarine may remain mechanically sound while the crew has ceased functioning as a team.

Morale is therefore a command concern, not a decorative luxury.

Crew members require:

- Fair pay.
- Clear instructions.
- Time to recover.
- Reasonable station leave.
- Recognition for good work.
- Consistent discipline.
- Adequate equipment.
- Honest communication.
- Some tolerance for harmless nonsense.
- A captain who does not treat every minor mistake as treason.

A vessel where nobody jokes is not necessarily professional.

It may simply be frightened.

Allow the crew to speak, laugh, purchase foolish hats, decorate their workspaces within reason, and occasionally do something harmless that would not appear in the official operating manual.

Harmless foolishness builds morale.

Armed foolishness builds medical workload.

The distinction should be maintained.

The captain's objective is to arrive with the mission complete and the crew still willing to sail again.

Arriving with everyone alive but actively debating the practicalities of throwing the captain out of the airlock represents only partial success.

XVI. CREW DISCIPLINE

Discipline beneath the ice is complicated by distance, mixed jurisdictions, private contracts, debts, political allegiances, criminal associations, union rules, and the traditional European belief that every regulation is optional if the nearest authority is several kilometres away.

Not every disagreement is mutiny.

Not every refusal is sabotage.

Not every criticism is insubordination.

The captain must distinguish between:

- A legitimate safety objection.
- A department reporting that an order is impossible.
- A worker raising a compensation issue.
- Refusal of ordinary duty without cause.
- Deliberate obstruction.
- Theft.
- Assault.
- Sabotage.
- An organized attempt to seize command.
- An assistant wearing a clown mask and behaving suspiciously, which is statistically inconclusive.

Discipline should be proportionate.

Possible responses include:

- Verbal correction.
- Reassignment.
- Removal from sensitive duties.
- Confiscation of weapons.
- Detention.
- Dismissal at the next station.
- Transfer to local authorities.
- Emergency force when the vessel or crew is under immediate threat.

Execution is not the standard punishment for incompetence.

If it were, Europa would have run out of submariners during the first decade.

XVII. STANDARD UNION 208

Because Even at Crush Depth, Someone Will Ask About Overtime

Submariner labor organization began almost as early as permanent settlement.

One of the earliest recorded disputes involved a Crampton-class cargo submarine supporting a prestigious government scientific mission.

The expedition's sponsors assumed the honor of serving science would compensate for long hours, dangerous transit, poor conditions, and miserable pay.

The crew concluded that honor could not be spent at the supply counter.

They refused departure, organized collectively, and demanded compensation proportionate to the voyage.

The dispute delayed the mission, embarrassed its sponsors, and produced one of Europa's earliest recognized submarine labor agreements.

From that tradition emerged Standard Union 208, now among the most widely recognized submariner organizations beneath the ice.

The union's influence differs between settlements, employers, and jurisdictions, but its standards are commonly referenced even aboard nonunion vessels.

Captains should familiarize themselves with the Standard Union 208 Standards of Operations, Submarine Fairness, and Crew Justice.

The standards generally recognize:

- Agreed pay scales.
- Hazard compensation.
- Clear duties.
- Reasonable rest.

- Necessary protective equipment.
- Transparent division of mission rewards.
- Medical treatment.
- Grievance procedures.
- Limits on arbitrary punishment.
- Compensation for extended duties.
- Restrictions on abandoning personnel.
- The radical principle that crew members should not be used as experimental subjects without asking first.

Union agreements may appear inconvenient.

They are generally less inconvenient than a reactor operator deciding mid-voyage that the collective bargaining session begins now.

XVIII. PAY AND COMPENSATION

Most crew members expect payment.

This continues to surprise certain employers.

A captain should establish:

- Base salary.
- Mission shares.
- Salvage distribution.
- Hazard bonuses.
- Equipment responsibility.
- Medical provisions.
- Personal spending rules.
- Permitted deductions.
- Conditions for renegotiation.

Do not promise money the vessel does not possess.

Do not alter shares after the work is complete because the mission cost more than expected.

Do not describe radiation exposure, gunfire, pressure trauma, parasitic infection, and prolonged monster attack as “valuable experience” in place of wages.

Experience is useful.

It is not accepted by most merchants.

A fairly compensated crew is more likely to:

- Follow orders.

- Report shortages.
 - Protect the vessel.
 - Remain employed.
 - Accept dangerous assignments.
 - Defend command.
 - Refrain from discussing whether the captain's chair would fetch a good price at the next station.
-

XIX. MID-MISSION PAY DISPUTES

Compensation disputes may arise during a voyage.

The mission may prove more dangerous than advertised. The route may become longer. Casualties may force surviving crew members to assume additional duties. An allegedly straightforward salvage contract may develop teeth, guns, infection, and an alien defense system.

The crew may request:

- Hazard pay.
- Increased mission shares.
- Compensation for additional duties.
- Payment for extended operations.
- Relief at the next station.
- Renegotiation before accepting another objective.

The captain should not immediately classify this as mutiny.

Listen.

Determine whether circumstances changed.

Consult department heads or representatives.

Consider the vessel's finances.

Negotiate while the complaint remains a complaint.

Replacing the entire engineering department at the next station generally costs more than paying the existing one enough to stop discussing industrial action beside the reactor.

When immediate payment is impossible, possible arrangements include:

- Deferred wages.
- Increased salvage shares.
- Priority payment from the next contract.
- Additional leave.
- Recorded debt.
- Mission bonuses.

- Release from contract.

A debt acknowledged may be tolerated.

A debt denied becomes a tradition, usually involving sabotage.

XX. RECRUITMENT

Recruitment determines how many future disciplinary hearings the captain will attend.

Crew members should be recruited through recognized stations, unions, agencies, formal contracts, or clear direct agreements.

Recruiting through criminal debt brokers, coercive labor arrangements, prison schemes, or suspicious individuals offering “fully trained reactor technicians” at half price may appear economical.

It commonly produces personnel who:

- Hold divided loyalties.
- Are wanted by several organizations.
- Lack the advertised skills.
- Conceal their identities.
- Resent command.
- Believe mutiny is a reasonable method of resignation.

Recruit above board.

Pay a known rate.

Explain the risks.

Record the agreement.

A captain who recruits through deception has selected deception as the vessel's founding policy.

XXI. JUSTICE ABOARD SHIP

There is no single universal justice system across Europa.

Coalition authorities, separatist settlements, corporations, unions, independent outposts, and criminal organizations all possess their own interpretations of law.

Many of these interpretations disagree violently.

Every captain should therefore establish onboard rules before someone tests them.

The crew should know:

- Who may order detention.
- Who may carry weapons.
- What constitutes sabotage.
- How theft is handled.
- How complaints are raised.
- How disputes are reviewed.
- When station authorities become involved.
- What emergency powers command may exercise.
- Who succeeds the captain.

Justice should not depend entirely on whether the captain has eaten recently.

Rules applied only to unpopular crew members are not discipline.

They are favoritism with paperwork.

Serious incidents should, where possible, involve several officers. Security may investigate. Medical may assess injuries or impairment. Department heads may provide testimony.

The captain makes the final shipboard decision.

They should avoid deciding the outcome first and then sending security to locate a reason.

XXII. MUTINY

Mutiny is an organized attempt to unlawfully seize, remove, confine, or override command.

It is not:

- A disagreement.
- A safety report.
- A wage grievance.
- An engineer saying the reactor cannot produce the requested power.
- A medic refusing to distribute morphine as a recreational beverage.
- A mechanic asking the captain to stop driving into walls.

Mutiny may involve:

- Seizing the helm.
- Detaining the captain.
- Disarming loyal personnel.
- Taking control of weapons.
- Sabotaging command systems.
- Issuing contradictory orders.
- Preventing completion of the mission.

- Organizing to replace lawful command by force.

The captain's response should reflect the danger.

Verbal disputes may still be negotiated.

Armed seizure of the helm may require security intervention.

Remember that most mutinies begin long before anyone reaches the command room.

They begin with:

- Unpaid wages.
- Reckless orders.
- Repeated humiliation.
- Ignored safety concerns.
- Arbitrary punishment.
- Unequal treatment.
- The conviction that lawful complaints will never be heard.

The most effective method of preventing mutiny is competent, fair command maintained before anyone begins quietly counting the weapons in the armory.

XXIII. COMMAND SUCCESSION

Every submarine requires a known order of succession.

A typical order may be:

1. Executive officer.
2. Relief captain or navigator.
3. Security chief.
4. Chief engineer.
5. Senior surviving department head.
6. Whoever can reach the helm without tripping over the previous candidates.

The exact order is less important than having one.

A submarine should not descend toward crush depth while five officers conduct a constitutional debate over whose hat is most captain-like.

The acting captain should announce:

- That command has transferred.
- Why it transferred.
- Whether the original captain is alive.
- Whether the transfer is temporary.

- What the vessel is doing next.

When the captain recovers, authority should normally be returned unless medical, disciplinary, or legal circumstances indicate otherwise.

XXIV. THE CAPTAIN AND THE HELM

The captain has primary authority over the helm.

The captain should not be the only person capable of using it.

Train a relief navigator.

Explain the route.

Teach basic docking.

Ensure another officer can hold depth, stop the vessel, and avoid large stationary objects.

A captain who hides all navigational knowledge to protect their authority may succeed until they become unconscious.

At that point, the submarine inherits both the captain's secrets and a wall.

Authority supported by crew competence is stronger than authority protected through ignorance.

XXV. THE CAPTAIN DURING A CRISIS

When several emergencies occur at once, establish priorities.

A useful order is:

1. Prevent immediate destruction.
2. Arrest uncontrolled descent or collision.
3. Stabilize the reactor.
4. Control catastrophic flooding.
5. Repel boarders.
6. Preserve breathable compartments.
7. Recover critical casualties.
8. Restore propulsion.
9. Reestablish department control.
10. Decide whether the mission still deserves to exist.

The captain should avoid leaving the helm to personally repair the first damaged item reported.

The captain's value lies in seeing the entire crisis.

If command abandons the helm to repair one junction box, nobody may remain to notice that the submarine is approaching crush depth, the away team is still outside, and the repair in question supplies the vending machine.

Delegate.

Monitor.

Reassign.

Intervene personally only when necessary.

A captain carrying a wrench may feel useful.

A captain maintaining control is useful.

XXVI. WHEN TO RETREAT

Retreat may become necessary when the vessel faces:

- Severe ammunition depletion.
- Multiple critical casualties.
- Reactor instability.
- Uncontrollable flooding.
- Navigation failure.
- Loss of medical capability.
- Overwhelming hostile forces.
- Insufficient fuel.
- Activated alien defenses.
- Heavy away-team losses.
- Damage beyond the crew's repair capacity.

Retreat does not mean the mission is lost forever.

It means the submarine remains available to fail again under better circumstances.

Preserving the crew and vessel provides another opportunity.

Continuing until everyone dies provides a wreck for someone else to salvage.

FINAL DUTY OF THE CAPTAIN

The captain's authority is broad because their responsibility is broader.

They are responsible for:

- The route.
- The mission.
- The department assignments.
- The crew roster.
- The away teams.
- The fuel.
- The ammunition.
- The medical reserves.
- The engineering supplies.
- The discipline.
- The pay.
- The succession plan.
- The decision to proceed or retreat.
- The general likelihood that anyone aboard will agree to another voyage.

A captain need not be the finest engineer, physician, gunner, mechanic, or diver.

They must know enough to ask each department the correct questions.

They must understand enough to recognize when the answer is alarming.

Keep the vessel moving.

Keep the departments informed.

Keep the crew paid.

Keep the sonar useful.

Keep the creatures outside.

Keep track of everyone entrusted to your command.

The captain does not own the crew.

The captain holds their survival in trust.

Command is not proven by how many people obey when security is standing behind you.

It is proven by how many crew members, after surviving the voyage and receiving their wages, willingly step aboard when the docking clamps release again.

Preferably without first demanding hazard pay.

Though, under Standard Union 208, they are entirely within their rights to ask.

CONCLUSION

The submariner's trade is not built upon heroic individuals acting alone.

It is built upon ordinary people performing necessary work under extraordinary pressure.

The most valuable crew member is not always the finest shot, the most gifted engineer, the quickest medic, or the boldest diver. It is often the person who can be trusted to carry out an order, report a problem, conserve supplies, help a struggling crewmate, and keep their judgment when the alarms begin.

Do not operate equipment you do not understand.

Do not take supplies you do not require.

Do not leave the vessel without authorization.

Do not use weapons without identifying the target and what lies beyond it.

Do not interfere with the helm or chain of command for personal satisfaction.

Do not turn mistakes into feuds.

Do not make Europa's work easier.

Learn your station. Maintain your equipment. Answer the radio. Watch the hatches. Return what you borrow. Help recover the wounded. Keep the decks clear. Respect the people whose lives are sealed inside the same hull as yours.

Beyond the glass is darkness, pressure, cold, radiation, and hunger.

Inside the hull is the crew.

Treat that distinction as sacred.

APPENDIX: STANDARD UNION 208

Mariners' Crew Rights, Submarine Labor Law, and Mutual Support Beneath the Ice

Issued under the authority of Standard Union 208, European Submariners' Mutual Aid and Labor Compact

Recognized across Coalition, separatist, corporate, independent, frontier, and irregularly administered waters

FOREWORD

No Vessel Sails Alone

Every submarine operating beneath the ice of Europa depends upon more than its captain, crew, reactor, and pressure hull.

It depends upon dockworkers who load its cargo.

It depends upon fabricators who produce its parts.

It depends upon miners who recover its fuel stock.

It depends upon station personnel who maintain docking arms, pumps, atmosphere, communications, and navigation beacons.

It depends upon medical workers who receive its wounded.

It depends upon merchants willing to sell it food, oxygen, ammunition, tools, medicine, batteries, welding fuel, and reactor rods.

It depends upon other submariners who answer distress calls in waters where no government vessel is likely to arrive in time.

This interdependence is the foundation of Standard Union 208.

The union's central principle is simple:

No mariner beneath the ice survives entirely by their own hand.

The most heavily armed captain still needs fuel.

The wealthiest station still needs food.

The most feared cartel still needs spare parts.

The most isolated outpost still requires someone willing to pilot a cargo vessel through the dark.

Europa has many governments, several currencies, dozens of legal systems, and more self-appointed authorities than it has reliable toilets. Yet nearly all of them have learned to recognize Standard Union 208 labor protections.

Some recognize them because they believe in fair treatment.

Some recognize them because they respect tradition.

Some recognize them because their insurers require it.

Some recognize them because the last administrator who ignored the union spent three months attempting to operate an outpost without food deliveries, replacement oxygen membranes, medical stock, or anyone willing to repair the docking collar.

The motivation is less important than the result.

Do not trifle with Union 208.

The sea is cold, the routes are long, and stations that lose the cooperation of mariners tend to die surprisingly quickly.

I. WHAT STANDARD UNION 208 IS

Standard Union 208 is the largest and oldest broadly recognized submariners' labor organization operating beneath the European ice.

It represents, supports, or provides contractual standards for:

- Captains.
- Navigators.
- Engineers.
- Mechanics.
- Medical officers.
- Security personnel.
- Gunnery crews.
- Cargo handlers.
- Salvage divers.
- Miners.
- Researchers assigned to submarine crews.
- Assistants and general hands.
- Dockworkers.
- Shuttle pilots.
- Station logistics personnel.
- Rescue and recovery teams.
- The numerous unfortunate specialists whose official job description ends with "and other duties as required."

The number 208 originally referred to the registration designation of the earliest recognized mutual-assistance compact under which the organization operated.

The designation survived long after the original registry ceased to matter.

Several later administrations attempted to rename the union.

The proposed titles included:

- European Maritime Labor Federation.
- Subaquatic Transit Workers' Association.
- United Under-Ice Operations Guild.
- Coalition-Recognized Submersible Personnel Compact.
- Interstation Nautical Employment Standards Authority.

None survived ordinary conversation.

Submariners continued calling it Union 208, and eventually the administrators stopped wasting paper.

The union is not a government, though it frequently performs duties that governments have neglected.

It is not a navy, though it has coordinated convoys, rescues, evacuations, and defensive actions.

It is not a commercial company, though it manages employment registries, benefit funds, emergency contracts, and supply guarantees.

It is not a criminal organization, though several station authorities have described it as one immediately before requesting union assistance.

It is, at its core, an agreement among working mariners that no crew should be forced to face Europa alone, unpaid, unequipped, abandoned, or without recourse.

II. THE CONDITIONS THAT CREATED THE UNION

The earliest years of permanent settlement beneath Europa's ice were defined by distance, optimism, desperation, and accounting departments located safely elsewhere.

Terran governments and scientific institutions viewed submarine crews primarily as transportation personnel.

The submariners viewed themselves as the people doing the dying.

Early crews were expected to carry scientists, equipment, fuel, biological samples, construction materials, and experimental machinery between isolated outposts. Contracts were often written by administrators who had never traveled beneath the ice and who considered danger an abstract category suitable for a liability waiver.

Pay was poor.

Food was inconsistent.

Protective equipment was reused far beyond its intended service life.

Submarines sailed with damaged pumps, depleted medical cabinets, unreliable reactors, and pressure hulls whose inspection records were optimistic works of fiction.

When crews protested, they were told that they were participating in a historic endeavor.

History, the crews discovered, did not pay for replacement limbs.

The early settlement authorities depended upon mariners but treated them as interchangeable labor. A crew lost in transit could be replaced from a growing pool of desperate colonists, prisoners, debtors, contractors, and workers seeking passage between stations.

For a time, this system functioned.

Then expansion increased the distances between stations.

Cargo volumes grew.

Routes became more dangerous.

Experienced submariners became more valuable.

The settlements learned that a trained reactor operator could not be replaced by handing a manual to a dockworker and wishing them luck.

Mariners began exchanging information about employers, pay rates, unsafe vessels, dishonest captains, and stations that refused to honor contracts.

These informal conversations became lists.

The lists became shared registries.

The registries became collective agreements.

The agreements became Standard Union 208.

III. THE CRAMPTON REFUSAL

The event most commonly identified as the union's beginning occurred aboard an early Crampton-class cargo submarine assigned to support a scientific mission between two of Europa's first permanent stations.

The mission was prestigious.

Its sponsors included a major Terran government, several scientific institutes, and enough senior officials to ensure that every document described the voyage as historically significant.

The crew's compensation was a pittance.

The sponsors argued that the journey was routine and that participation in an important scientific mission carried value beyond wages.

The crew reviewed the planned transit.

They noted the distance.

They noted the poor state of the vessel.

They noted the inadequate medical supplies.

They noted that the route passed through waters where two previous crews had disappeared.

They noted that the contract contained generous protections for scientific equipment and almost none for the people transporting it.

The crew refused to depart.

The station director threatened dismissal.

The crew replied that dismissal would still leave the submarine at the dock.

Replacement personnel were sought.

Every qualified submariner on the station declined.

Cargo handlers refused to load the scientific equipment.

The reactor technicians refused to certify the vessel.

Dockworkers refused to release the clamps.

The mission's sponsors discovered that owning a submarine and commanding a submarine to sail were not the same thing.

Negotiations followed.

The crew secured:

- Increased wages.
- Hazard compensation.
- Improved medical supplies.
- A minimum fuel reserve.
- Written rescue obligations.
- Payment to families in the event of death.
- Recognition of a crew representative.
- The right to refuse departure aboard an uncertified vessel.

The submarine completed the mission.

The scientific institutions received their samples.

The government received its public triumph.

The crew received wages sufficient to make the risk something other than institutional theft.

Copies of the agreement spread between stations.

The clauses became known informally as the Crampton Standards.

They later formed the first body of Standard Union 208 contract law.

IV. THE FIRST CHARTER

The first Union 208 charter was not written in a courtroom or legislative chamber.

It was assembled from voyage contracts, handwritten wage agreements, rescue promises, crew manifests, dockworker pledges, and notes passed between submarines.

The charter recognized several principles that now appear obvious but were revolutionary at the time:

1. A crew member is entitled to know the work expected of them.
2. A crew member is entitled to know how they will be paid.
3. A submarine must carry adequate fuel, oxygen, medical supplies, and repair equipment for its planned route.
4. A captain must not knowingly abandon crew members without necessity.
5. A worker may refuse an obviously suicidal order when no immediate greater danger requires compliance.
6. Injured crew members are entitled to treatment.
7. Families and designated dependents are entitled to agreed death benefits.
8. Wage disputes must not be settled through summary execution.

The eighth provision was included because several employers had apparently found the matter unclear.

The charter also established obligations upon crew members.

Union protections did not excuse sabotage, theft, assault, deliberate dereliction, or refusal to perform ordinary agreed duties.

The relationship was reciprocal.

Captains owed fairness.

Crews owed service.

Employers owed payment.

The union owed support.

V. TRIALS, SUPPRESSION, AND THE BLACKLIST YEARS

The growth of Union 208 was not peaceful.

Corporate operators regarded collective bargaining as interference with commercial freedom.

Station administrators described union action as obstruction.

Private security companies treated organizers as agitators.

Several early union representatives disappeared during transit.

Others were arrested on charges that ranged from contractual interference to sedition, theft of company time, illegal assembly, and failure to respect management authority.

Employers established blacklists.

A worker identified as a union organizer could be denied employment across several stations.

Captains who recognized union standards were refused contracts.

Merchants were pressured not to supply vessels carrying union representatives.

In response, the union developed its own employment and supply network.

Union captains hired blacklisted workers.

Union dockworkers prioritized compliant vessels.

Union-affiliated merchants extended credit.

Mechanics shared parts.

Medical officers created emergency treatment funds.

Submarines carried messages between isolated chapters.

The attempt to starve the union produced the mutual-aid system that ultimately made it far more powerful.

The blacklists failed because they assumed workers required access to employers more than employers required access to workers.

On Europa, this assumption has a limited operational lifespan.

A corporation may own twelve submarines.

Without crews, it owns twelve expensive pressure vessels attached to a dock.

VI. THE UNION MUTUAL-AID NETWORK

Standard Union 208 is held together not merely by contracts but by material support.

Union chapters maintain or coordinate:

- Emergency food reserves.
- Medical relief.
- Replacement equipment.
- Temporary housing.
- Legal and arbitration assistance.
- Rescue funds.
- Survivor benefits.
- Crew-transfer registries.
- Distress-call relays.
- Blacklists of dangerous vessels and employers.
- Safe ports.
- Fuel loans.
- Shared repair stock.
- Burial and recovery services.
- Wage-guarantee accounts.
- Assistance for stranded or abandoned mariners.

A submariner in good standing may arrive at a distant station with little more than identification, a work record, and a story involving an unfortunate ballast incident.

The local chapter may help them find work, secure temporary shelter, obtain replacement clothing, contact family, or pursue unpaid wages.

This support is not charity in the ordinary sense.

It is collective insurance maintained by mariners who understand that any of them may one day emerge from an airlock into a station where their employer has vanished and their submarine has departed without them.

The union's strength lies in the knowledge that today's rescue fund may become tomorrow's fuel loan, survivor payment, or emergency meal.

VII. THE SAVILLE SACRIFICE

No history of Standard Union 208 is complete without the crew of the *Saville*.

The *Saville* was a medium cargo submarine operating along a frontier supply route serving several mining stations and one heavily fortified commercial enclave controlled by the Harrow Deep Combine.

The Combine was not legally recognized as a government.

It possessed weapons, territory, debt contracts, prisons, security personnel, and a tax office.

On Europa, this distinction is often considered academic.

The *Saville* carried food, replacement oxygen membranes, reactor components, medical supplies, and pressure-seal materials.

Its crew operated under a Union 208 contract.

When the submarine reached the Combine's territory, cartel representatives ordered the captain to accept a revised contract.

The revised terms reduced crew pay, transferred all salvage rights to the Combine, permitted indefinite extension of the voyage, and allowed armed supervisors to discipline crew members without appeal.

The captain refused.

The crew supported the refusal.

The Combine impounded the vessel and detained twelve members of the crew.

Union negotiators demanded their release.

The Combine responded that union law did not apply within its territory.

The union then issued what became known as the Saville Notice.

No Union 208 vessel would deliver cargo to the enclave.

No union dockworker would service its submarines.

No affiliated mechanic would certify its pressure systems.

No union medical officer would contract with Combine transports.

No merchant protected by union convoy agreements would carry restricted supplies into its territory.

The Combine believed it could replace the union.

It hired independent captains.

Several accepted payment, collected cargo, and declined to complete delivery after learning the route had lost rescue coverage.

Others discovered that no qualified dockworkers would load their vessels.

Nonunion repair crews attempted to maintain the enclave's submarines and produced an impressive series of electrical fires.

Food became scarce.

Reactor parts failed.

Medical supplies dwindled.

The enclave's station pumps began operating without reliable replacement components.

The Combine retaliated against the detained *Saville* crew.

Several were beaten.

Several were subjected to forced labor.

Four died in custody.

The surviving crew members continued refusing the revised contract.

They were offered individual release in exchange for renouncing the union and accepting personal employment under the Combine.

None accepted.

After thirty-seven days, the enclave's outer atmosphere-processing array suffered a major failure.

The Combine requested emergency parts.

Union 208 replied that humanitarian oxygen support would be delivered for civilians, but no commercial service would resume until the surviving crew were released, the bodies of the dead were returned, and the original contracts were honored.

The Combine yielded.

The surviving *Saville* crew left under union escort.

The dead were carried aboard four submarines, each escorted by another vessel with its lights extinguished.

Every union vessel along the route sounded one controlled sonar pulse as the convoy passed.

The pulse later became part of the annual *Saville* observance.

The four dead became known as the *Saville* Crewmen, though later memorials often honor all twelve detainees under the same title.

Their sacrifice established a principle now understood throughout Europa:

An attack upon a union crew may be answered not by gunfire, but by absence.

No food.

No fuel.

No medicine.

No parts.

No crews.

No deliveries.

The Saville settlement was recognized not only by Coalition authorities and legitimate station governments but by cartels, pirate enclaves, separatist commands, smugglers, and private security organizations.

They did not all develop respect for labor rights.

They developed respect for logistics.

That has proven sufficient.

VIII. THE SAVILLE CLAUSE

Modern Union 208 agreements contain a provision commonly called the Saville Clause.

It states that deliberate detention, torture, disappearance, execution, forced labor, or coercive contract alteration affecting a protected crew may trigger collective suspension of services.

The scope of such a suspension may include:

- Cargo refusal.
- Docking refusal.
- Repair refusal.
- Crew refusal.
- Fuel embargo.
- Suspension of convoy protection.
- Withdrawal of rescue guarantees.
- Refusal to recognize local employment contracts.
- Public listing of the offending station or organization.
- Free passage for civilians fleeing the affected territory.
- Humanitarian deliveries administered independently of the offending authority.

The union distinguishes between pressure upon an abusive authority and harm to ordinary station residents.

Food and oxygen may still be provided directly to civilians.

Medical evacuations may continue.

Emergency rescue may be rendered.

The purpose of a Saville action is not to kill a station.

It is to demonstrate how quickly a station's leadership can lose control when the labor supporting it withdraws cooperation.

This distinction is important.

It is also frequently inconvenient for station leadership, which is the point.

IX. WHY CARTELS RECOGNIZE UNION 208

Criminal organizations do not generally recognize laws because they are impressed by their moral clarity.

They recognize consequences.

A cartel may ignore a court ruling.

It may bribe an inspector.

It may assassinate an administrator.

It may threaten a captain.

It cannot easily manufacture experienced submariners, reactor technicians, pressure-hull welders, surgeons, and cargo pilots on demand.

Criminal stations are often more dependent upon maritime labor than lawful ones.

They cannot openly purchase certain goods.

They rely upon smugglers.

They need pilots familiar with hidden routes.

They require mechanics willing to work on modified vessels.

They require medical officers who ask few questions.

They require merchants willing to enter dangerous territory.

A cartel that violates Union 208 rules may discover that all of these people have independently decided to be elsewhere.

Even pirates observe certain union protections.

They may seize cargo.

They may demand ransom.

They may rob a vessel.

But many will release ordinary crew members, honor medical neutrality, and avoid interfering with union relief shipments.

This restraint should not be mistaken for kindness.

It is informed self-preservation.

A pirate captain may steal ten crates of ammunition.

They will hesitate before creating a labor dispute that prevents every friendly station from selling them food.

X. STANDARD EMPLOYMENT CONTRACT LAW

Union 208 agreements form the basis of common submarine employment law across much of Europa.

Local jurisdictions may vary, but a valid maritime employment contract is generally expected to identify:

- The employer.
- The captain.
- The vessel.
- The crew member.
- The assigned profession.
- The expected duties.
- The base wage.
- Mission shares.
- Salvage rights.
- Hazard compensation.
- Contract duration.
- Expected route.
- Conditions for extension.
- Medical provisions.
- Death benefits.
- Disciplinary procedures.
- Grounds for termination.
- Conditions for resignation.
- The responsible jurisdiction.
- The process for dispute resolution.

A contract that omits all meaningful obligations while granting the employer unlimited authority is not considered “flexible.”

It is considered suspect.

The fact that a crew member signed a document does not automatically make every term enforceable.

Union arbiters routinely reject provisions that involve:

- Unpaid indefinite service.
- Debt inheritance.
- Forced medical experimentation.
- Unrestricted corporal punishment.
- Summary execution for minor violations.
- Confiscation of all personal salary.
- Forced abandonment.
- Waiver of rescue rights.
- Retroactive wage reduction.
- Unlimited extension without compensation.
- Ownership of a crew member's body after death.
- Mandatory participation in clown-related research.

The last provision was added after an arbitration panel determined that existing language was somehow not specific enough.

XI. THE RIGHT TO CLEAR TERMS

Every mariner has the right to understand the essential conditions of employment.

A captain or employer should state:

- What the vessel is doing.
- Where it is going.
- What the crew member will be paid.
- What dangers are reasonably expected.
- How mission rewards will be divided.
- What equipment will be provided.
- What conduct may result in discipline.
- When the contract ends.

No employer may rely upon deliberately obscure language to conceal dangerous duties.

A salvage contract is not automatically an assault contract.

A cargo voyage is not automatically a ruin expedition.

A station-transfer agreement is not automatically consent to remain aboard for six additional missions.

Operations may change unexpectedly.

Europa is not known for respecting paperwork.

When conditions materially change, compensation and consent may need to be renegotiated.

XII. THE RIGHT TO FAIR COMPENSATION

Submariners are entitled to the compensation agreed upon before departure.

This may include:

- Base wages.
- Mission completion pay.
- Salvage shares.
- Hazard bonuses.
- Overtime or extended-voyage pay.
- Specialist pay.
- Death or injury benefits.
- Recruitment bonuses.
- Retention bonuses.
- Compensation for assuming additional duties.
- Emergency rescue awards.

Wages may not be reduced after the work has been completed merely because the voyage proved less profitable than expected.

The crew does not become an involuntary investor whenever the captain makes a poor commercial decision.

Likewise, a captain may not promise extraordinary shares during an emergency and later dismiss the promise as something said under stress.

Most important statements aboard a submarine are made under stress.

That does not make them imaginary.

XIII. HAZARD PAY

Hazard is an ordinary feature of submarine work.

Extraordinary hazard is a separate matter.

Hazard compensation may be required for:

- Entering known hostile nests.
- Boarding enemy vessels.
- Exploring active alien ruins.
- Handling unstable artifacts.
- Working in radiation.

- Clearing severe biological contamination.
- Operating beyond agreed safe routes.
- Continuing after major loss of medical capability.
- Performing duties outside the original contract.
- Entering waters subject to active military conflict.
- Working with defective safety equipment.
- Recovering bodies from highly dangerous environments.

The captain should not describe every danger as ordinary merely because danger is common.

There is a meaningful difference between routine submarine risk and being ordered to carry a heat-emitting alien device through a compartment full of volatile ammunition.

Union arbiters are familiar with the distinction.

XIV. THE RIGHT TO NECESSARY EQUIPMENT

A crew member must be provided with equipment reasonably required to perform assigned duties.

This includes, where applicable:

- Functional diving suits.
- Oxygen.
- Repair tools.
- Medical supplies.
- Protective clothing.
- Weapons.
- Ammunition.
- Radiation protection.
- Secure communications.
- Lighting.
- Access to appropriate workspaces.
- Safe storage.
- Emergency escape equipment.

An employer may require reasonable care of issued equipment.

They may charge for deliberate theft or reckless destruction under an agreed procedure.

They may not require a miner to enter open water without oxygen and later classify the resulting death as a performance issue.

Equipment must be functional, not merely present.

A diving suit with a missing seal is not protective equipment.

It is a burial garment with sleeves.

XV. THE RIGHT TO FOOD, WATER, AIR, AND HABITABLE CONDITIONS

Every crew member is entitled to the basic materials required for life.

A vessel must maintain, within reasonable operational limits:

- Breathable atmosphere.
- Potable water or equivalent supply.
- Adequate food.
- Sanitary facilities.
- Safe sleeping arrangements when voyages require extended rest.
- Protection from unnecessary radiation.
- Reasonable temperature control.
- Access to emergency medical care.

The union recognizes that combat and disaster may temporarily make conditions intolerable.

It does not recognize deliberate deprivation as ordinary management.

Food may not be withheld as punishment.

Oxygen may not be restricted to encourage productivity.

Medical treatment may not be conditioned upon obedience unrelated to the treatment.

A captain attempting to motivate the crew by disabling the oxygen generator will generally obtain increased movement.

Most of it will be toward the captain.

XVI. THE RIGHT TO MEDICAL CARE

Injured or ill crew members are entitled to timely treatment consistent with available resources.

Medical personnel must prioritize according to clinical need, not rank, wealth, popularity, or ability to threaten the doctor.

The captain is entitled to command consideration where survival of the vessel genuinely depends upon their recovery.

This does not mean a captain with a bruised wrist takes priority over an engineer suffering catastrophic blood loss.

Union medical law also recognizes:

- Confidential handling of personal medical information.
- Protection from involuntary experimentation.
- The right to refuse nonemergency treatment.
- The duty to report contagious or vessel-threatening conditions.
- Access to pain treatment when medically justified.
- Protection against punitive withholding of medication.
- Compensation for occupational injury.

Medical treatment is not a privilege granted for good behavior.

Neither is the medical cabinet a public narcotics buffet.

The distinction is addressed elsewhere in this guide and should not require another incident involving Jerry from engineering.

XVII. THE RIGHT TO REFUSE MANIFESTLY SUICIDAL ORDERS

A crew member may refuse an order when:

- The order presents an immediate and extreme danger.
- The danger is not reasonably necessary to preserve the vessel or crew.
- The crew member lacks required equipment or training.
- The order serves no legitimate operational purpose.
- Compliance would violate a recognized humanitarian protection.
- The order requires deliberate criminal harm unrelated to survival.

This right is narrow.

It does not permit refusal of every unpleasant, dangerous, or inconvenient task.

Submarine work is inherently dangerous.

A mechanic may still be ordered to repair a breach during an attack.

A security officer may still be ordered to confront hostile boarders.

A medic may still be required to enter a damaged compartment to reach casualties.

The relevant question is whether the danger is necessary, proportionate, and connected to preserving the vessel or completing agreed work.

“Go outside without a suit because I am curious what happens” is not a lawful order.

“Repair the external docking mechanism before we are crushed against the station” may be.

XVIII. THE RIGHT OF RETURN

Crew members sent outside the submarine have a right to reasonable recovery.

The captain must not knowingly abandon an away team while recovery remains practical and does not create an overwhelming danger to the vessel.

This does not require a captain to sacrifice the entire submarine for one missing diver.

It requires:

- Maintaining an accurate roster.
- Establishing recall procedures.
- Allowing reasonable return time.
- Attempting communication.
- Considering rescue.
- Recording why departure became necessary.
- Reporting abandoned personnel at the earliest safe opportunity.

Deliberately leaving a disliked crew member outside is not command discretion.

It is attempted murder with navigational elements.

XIX. THE RIGHT TO RESCUE

Union 208 vessels are expected to respond to distress calls when reasonably capable.

The duty is affected by:

- Distance.
- Fuel.
- Current mission.
- Crew condition.
- Threat level.
- Vessel capability.
- Whether responding would create a second casualty.

A submarine need not destroy itself attempting an impossible rescue.

It should not ignore a viable rescue merely because no payment has been guaranteed.

Union rescue obligations helped create the under-ice routes as a functioning network rather than a collection of isolated commercial risks.

Mariners aid mariners because Europa is too large, too dark, and too hostile for anyone to assume another vessel will answer.

Rescue compensation may be negotiated afterward.

Breathing should begin first.

XX. THE RIGHT TO SHORE LEAVE AND REST

Crew members are entitled to reasonable rest and shore leave when operational circumstances permit.

This does not mean every station call must become a three-day festival.

It means crews cannot be expected to operate indefinitely without recovery.

Fatigued workers:

- Make mistakes.
- Miss threats.
- Mismanage reactors.
- Misdiagnose patients.
- Fire through teammates.
- Forget fuel.
- Fall asleep at the helm.
- Become increasingly receptive to mutiny as a form of recreation.

The captain may recall personnel for departure or emergency duty.

Repeated denial of rest without necessity may trigger grievance, compensation claims, or refusal of additional missions.

A crew member who has been awake for thirty hours is not displaying admirable dedication.

They are an equipment failure that has learned to speak.

XXI. THE RIGHT TO REPRESENTATION

A union crew may select or recognize a representative.

The representative may:

- Present grievances.
- Review contract terms.
- Participate in disciplinary proceedings.
- Negotiate compensation.
- Communicate with station chapters.
- Request arbitration.

- Document unsafe conditions.
- Coordinate collective action.
- Ensure benefits reach injured workers or dependents.

The representative is not a second captain.

They do not direct navigation or countermand emergency orders merely because they possess a union badge.

The captain commands the vessel.

The representative protects the agreement under which the crew serves.

Competent captains understand that these roles can coexist.

Incompetent captains generally learn through arbitration.

XXII. THE RIGHT TO GRIEVANCE

Crew members may raise concerns about:

- Unpaid wages.
- Unsafe equipment.
- Discriminatory treatment.
- Contract violations.
- Improper punishment.
- Withheld medical care.
- Abandonment.
- Unfair salvage distribution.
- Coercive extension of service.
- Retaliation.
- Inadequate food or oxygen.
- Unlawful orders.
- Misrepresentation during recruitment.

A grievance should be reported through the vessel's established process when possible.

This may involve:

1. Reporting to the department head.
2. Reporting to the captain.
3. Consulting the union representative.
4. Recording the complaint.
5. Requesting review at the next station.
6. Seeking emergency arbitration when delay would cause serious harm.

The union encourages resolution aboard the vessel when practical.

It does not require workers to continue using an internal process controlled entirely by the person accused of the violation.

XXIII. DISCIPLINE AND DUE PROCESS

Union rules do not prevent discipline.

They prevent arbitrary discipline.

Crew members may be corrected or punished for:

- Sabotage.
- Assault.
- Theft.
- Deliberate insubordination.
- Unauthorized weapons use.
- Reckless endangerment.
- Abandoning duty.
- Misuse of medical supplies.
- Tampering with the reactor.
- Unauthorized movement of the submarine.
- Repeated refusal of agreed work.
- Bringing an unidentified organism aboard and placing it in the ballast tank.

Discipline should be proportionate to the violation.

Possible measures include:

- Warning.
- Retraining.
- Reassignment.
- Loss of sensitive privileges.
- Temporary detention.
- Restitution.
- Dismissal.
- Transfer to station authorities.
- Emergency force when an immediate threat exists.

A crew member should be informed of the reason for serious discipline.

Evidence should be preserved when possible.

Witnesses should be consulted.

The accused should be permitted to respond unless the vessel is in immediate danger.

Summary execution because someone “looked suspicious” is generally disfavored.

This remains true even when the person is dressed as a clown.

XXIV. MUTINY, WORK STOPPAGE, AND COLLECTIVE REFUSAL

Union action is not automatically mutiny.

Mutiny seeks unlawful control of the vessel.

A labor action seeks enforcement or renegotiation of employment terms.

The distinction may become difficult when the reactor operator folds their arms beside the control panel, but it remains important.

A protected collective refusal may arise when:

- Wages have not been paid.
- The vessel is manifestly unsafe.
- Essential equipment is unavailable.
- The crew is ordered beyond agreed duties without negotiation.
- The captain refuses to address a serious contract violation.
- A union representative has been unlawfully detained.
- The employer attempts forced labor.

A work stoppage must not deliberately destroy the vessel or abandon innocent personnel to immediate danger.

Essential life-support functions should be preserved where possible.

Crew members may refuse to depart from port more safely than they can refuse to repair an active hull breach.

The ocean does not suspend pressure while labor and management resolve their differences.

Union law expects both judgment and solidarity.

XXV. SALVAGE RIGHTS

Salvage is a common source of dispute.

Contracts should specify:

- Whether salvage belongs to the vessel, employer, or crew.
- What percentage is allocated to operating costs.
- How individual discoveries are treated.

- Whether personal items may be claimed.
- How artifacts are valued.
- How mission-required objects differ from discretionary salvage.
- Whether dangerous recovery earns additional compensation.

A captain should not declare all valuable objects ship property while assigning all worthless objects to the crew.

Neither should crew members conceal high-value salvage in personal lockers and expect the union to defend the practice.

Transparent division prevents the traditional portside argument in which everyone suddenly remembers a different version of the agreement.

XXVI. INJURY, DISABILITY, AND DEATH BENEFITS

Submarine labor carries a high risk of permanent injury.

Union agreements may provide:

- Treatment costs.
- Recovery wages.
- Disability compensation.
- Prosthetic assistance.
- Transportation to suitable medical facilities.
- Retraining for another profession.
- Support for dependents.
- Funeral or recovery expenses.
- Death benefits.
- Payment of wages owed at the time of death.

The loss of a crew member does not erase the employer's obligations.

A captain may not classify the dead as having resigned.

A corporation may not withhold wages because the worker failed to personally collect them.

The union maintains beneficiary registries so that compensation can reach families, partners, children, or other designated dependents across distant stations.

Where no beneficiary exists, agreed funds may support crew memorials, recovery operations, or mutual aid.

XXVII. DEBT AND FORCED LABOR

Debt is common on Europa.

Forced labor disguised as debt repayment is also common.

Union 208 recognizes legitimate employment advances and repayment agreements.

It does not recognize:

- Infinite interest.
- Inherited employment debt.
- Debt created through employer-controlled prices.
- Charges for basic air.
- Charges for mandatory safety equipment.
- Retroactive recruitment fees.
- Debt extended through punishment.
- Forced continuation of service after repayment.
- Sale of a crew member's contract without consent.

A mariner who owes money remains a person, not cargo.

Criminal organizations frequently disagree.

They nevertheless tend to respect union intervention after being reminded how many debtors operate their cargo routes.

XXVIII. CONTRACT EXTENSION

A captain may request that a crew extend service beyond the original contract.

The crew may agree.

Additional compensation should be established.

Emergency circumstances may require temporary extension when immediate departure would endanger the vessel or leave the crew stranded.

Such necessity does not grant indefinite ownership of the crew's labor.

Once a safe station is reached, workers must be allowed to renegotiate, transfer, or end service according to the agreement.

The phrase "one more mission" has been used to justify voyages lasting several months.

Union contracts therefore recommend replacing it with an actual number.

XXIX. CAPTAIN'S RIGHTS UNDER UNION 208

Union law protects captains as workers and commanding officers.

Captains are entitled to:

- Agreed compensation.
- Clear ownership or employment terms.
- Accurate crew qualifications.
- Protection against sabotage.
- Authority to direct ordinary vessel operations.
- Emergency command powers.
- Disciplinary support for serious misconduct.
- Arbitration with employers.
- Protection from being forced to operate an unfit vessel.
- Refusal of missions beyond contractual or mechanical capability.
- Crew cooperation with lawful orders.
- Union rescue and mutual aid.

Many captains are themselves union members.

The union does not treat command and labor as natural enemies.

A captain may be responsible for enforcing an employer's contract while also being underpaid by the same employer.

Europa is efficient in that way.

XXX. CREW OBLIGATIONS

Rights carry obligations.

Union-protected crew members must:

- Perform agreed duties.
- Follow lawful orders.
- Preserve ship property.
- Report hazards.
- Maintain reasonable competence.
- Avoid sabotage.
- Avoid theft.
- Avoid unnecessary violence.
- Conserve supplies.
- Observe safety procedures.
- Return from authorized leave.
- Assist during emergencies.
- Respect the rights of other crew members.

- Participate honestly in investigations and arbitration.

Union membership is not immunity from consequences.

The union will defend fair process.

It will not necessarily defend the conduct that created the process.

A member caught deliberately putting volatile compound into the reactor does not improve their case by presenting a union card.

XXXI. UNION ARBITRATION

Disputes that cannot be resolved aboard ship may be referred to Union 208 arbitration.

Arbitrators may be:

- Senior captains.
- Engineers.
- Medical officers.
- Labor delegates.
- Station representatives.
- Legal specialists.
- Experienced mariners recognized by both parties.

Arbitration reviews:

- Contracts.
- Logs.
- Wage records.
- Medical reports.
- Crew testimony.
- Security recordings.
- Voyage data.
- Supply records.
- Disciplinary actions.
- Evidence of changed mission conditions.

Remedies may include:

- Payment of wages.
- Restitution.
- Contract termination.
- Reinstatement.
- Blacklisting.
- Removal from a hiring registry.
- Compensation.

- Revised salvage distribution.
- Formal censure.
- Referral to station authorities.
- Collective action.

Union arbitration is often faster than station courts.

This is partly because union arbiters understand submarine work.

It is also because they do not require six months to determine whether a completely flooded reactor room was unsafe.

XXXII. THE RED REGISTRY

Union 208 maintains records of employers, captains, vessels, and stations accused or found responsible for severe violations.

This is commonly known as the Red Registry.

Listings may warn of:

- Unpaid wages.
- Unsafe vessels.
- Abandonment.
- Forced labor.
- Disappearance of crew.
- Fraudulent contracts.
- Repeated medical neglect.
- Violent discipline.
- Salvage theft.
- Union suppression.
- Failure to honor arbitration.
- Saville-level violations.

A registry listing may prevent an employer from recruiting experienced crew.

It may increase prices.

It may eliminate access to convoy protection.

It may cause dockworkers to refuse service.

The registry is not intended for personal revenge.

False reporting is itself a serious union offense.

A Red Registry entry must be supported by evidence, repeated credible complaints, or formal findings.

The system's power depends upon trust.

XXXIII. LOGISTICAL SANCTIONS

The union's greatest power is not military.

It is logistical.

A submarine cannot operate without:

- Fuel.
- Oxygen.
- Ammunition.
- Welding supplies.
- Food.
- Medicine.
- Parts.
- Repairs.
- Docking.
- Crew.
- Rescue coverage.

A station cannot survive without:

- Cargo.
- Replacement machinery.
- Medical shipments.
- Agricultural support.
- Power components.
- Skilled labor.
- Transport.
- Waste removal.
- Emergency evacuation.
- Trade access.

When a station violates fundamental union protections, the union may suspend cooperation.

This does not require every mariner to share the same political beliefs.

They need only understand that permitting one station to seize crews without consequence makes every crew less safe.

Today it is someone else's contract.

Tomorrow it is your engineer.

The day after, it is you.

Solidarity beneath the ice is not merely ideological.

It is preventative maintenance for civilization.

XXXIV. STATIONS AND THE CONSEQUENCES OF ISOLATION

A station may appear self-sufficient.

It has reactors.

It has hydroponics.

It has workshops.

It has armed security.

It has storage.

It may even have enough food for several months.

Then a circulation pump fails.

A crop fungus spreads.

A reactor component fractures.

A medical outbreak consumes antibiotics.

A docking seal becomes unreliable.

A critical specialist dies.

Without trade and maritime labor, small failures become cumulative.

A station denied regular support may survive for a time.

It will begin cannibalizing equipment.

It will ration.

It will postpone maintenance.

It will lose redundancy.

Eventually, a system fails that cannot be repaired with the parts available.

This is why even powerful organizations negotiate with Union 208.

Do not dare trifle with union rules under the assumption that guns make a station independent.

Guns do not fabricate oxygen membranes.

Guns do not repair food synthesizers.

Guns do not pilot supply submarines.

Guns can persuade a person to work briefly.

They are less effective at persuading an entire maritime network to deliver reactor parts.

XXXV. HUMANITARIAN NEUTRALITY

Union sanctions are not intended to punish civilians.

Union 208 maintains humanitarian exceptions for:

- Food.
- Oxygen.
- Emergency medicine.
- Civilian evacuation.
- Rescue.
- Epidemic control.
- Life-support repair.
- Child and dependent support.

Deliveries may be supervised to prevent seizure by the sanctioned authority.

Medical workers may enter under neutrality.

Civilian transports may receive union escort.

The union's objective is compliance, not starvation.

This principle distinguishes collective labor enforcement from siege warfare.

Organizations attempting to redirect humanitarian shipments for military or commercial use may face expanded sanctions.

Union personnel have a long memory for those who steal children's oxygen to refill a private attack boat.

XXXVI. THE SAVILLE OBSERVANCE

Each year, participating vessels observe the Saville sacrifice.

The exact date varies between station calendars, but the traditional observance includes:

- Reading the names of the detained crew.
- A moment of engine silence.
- One controlled sonar pulse.
- Inspection of crew contracts.
- Confirmation of beneficiary records.
- Contributions to the mutual-aid fund.
- Recognition of mariners lost during the previous year.

The sonar pulse represents the message carried along the route when the *Saville* dead were returned.

One pulse means:

We heard.

A second vessel may answer:

We remember.

The ceremony is brief.

Submariners generally prefer memorials that do not interfere excessively with departure schedules.

The meaning is not diminished by efficiency.

XXXVII. UNION HALLS

Most significant stations maintain a Union 208 office, desk, hall, or at minimum a person sitting beside a filing cabinet who insists that the filing cabinet constitutes an office.

A union hall may provide:

- Job postings.
- Contract review.
- Crew records.
- Benefit registration.
- Arbitration requests.
- Training notices.
- Distress reports.
- Blacklist information.
- Temporary lodging.
- Mutual-aid payments.
- Memorial records.
- Strong coffee.
- Opinions about captains who did not request them.

Union halls also function as information exchanges.

A captain seeking qualified crew may learn which engineers are available.

A mariner considering employment may learn whether the vessel has a history of returning with fewer people than it departed with.

This information saves lives.

It also explains why certain captains prefer recruiting in bars where employment records are less visible.

XXXVIII. TRAINING AND CERTIFICATION

Union 208 promotes basic standards of competence.

Training may include:

- Reactor operation.
- Hull repair.
- Medical triage.
- Diving safety.
- Weapons discipline.
- Navigation.
- Sonar use.
- Ballast management.
- Salvage procedures.
- Artifact handling.
- Crew leadership.
- Labor rights.
- Contract literacy.
- Emergency evacuation.

Certification is not universal.

Frontier conditions frequently produce competent workers without formal credentials and formally credentialed workers who should not be permitted near machinery.

The union therefore recognizes documented experience, supervised testing, and practical examination.

No certificate guarantees competence.

The absence of one does not prove incompetence.

The captain remains responsible for determining whether the person claiming to be a reactor expert knows which part of the reactor is the reactor.

XXXIX. UNION RELATIONS WITH GOVERNMENTS

Coalition authorities formally recognize many Union 208 standards.

Separatist governments recognize others.

Independent stations adopt portions suited to local conditions.

Corporate enclaves often create internal labor codes that resemble union rules while carefully avoiding the word "union."

This is considered progress.

Union 208 does not require political uniformity.

Its members include people loyal to competing governments, corporations, movements, settlements, and causes.

The common interest is maritime survival and fair employment.

Political conflicts may affect routes and contracts.

They do not erase the principle that crews deserve air, pay, equipment, treatment, and a reasonable chance of returning.

XL. UNION RELATIONS WITH CRIMINAL ORGANIZATIONS

Union 208 does not endorse criminal cartels.

It does negotiate with them.

This has produced criticism from authorities whose own rescue ships have occasionally declined to enter cartel waters.

The union's position is practical.

Mariners work everywhere.

If a crew is detained by a criminal organization, ideological disapproval alone will not free them.

Union intermediaries may negotiate:

- Crew release.
- Medical access.
- Return of bodies.
- Recognition of contracts.
- Safe passage.
- Ransom limitations.

- Protection of humanitarian cargo.
- Compliance with Saville provisions.

Some cartels maintain recognized union liaisons.

This arrangement is absurd.

It is also responsible for keeping many people alive.

Europa rewards institutions capable of functioning despite absurdity.

XLI. THE BLACK ICE PROVISION

Union agreements recognize that certain emergencies suspend ordinary procedures.

This is known as the Black Ice Provision.

During immediate threat to the vessel, the captain may:

- Reassign personnel.
- Restrict movement.
- Recall shore leave.
- Use emergency supplies.
- Delay nonessential grievances.
- Order dangerous work necessary to preserve life.
- Temporarily modify ordinary rest schedules.
- Detain individuals posing an immediate threat.

The Black Ice Provision does not permit:

- Theft of wages.
- Unrelated punishment.
- Indefinite forced labor.
- Execution without necessity.
- Concealment of contract violations.
- Permanent cancellation of union rights.

Emergency authority ends when the emergency ends.

Captains who remain under permanent emergency powers for six consecutive months are advised to admit that they have invented a dictatorship.

XLII. THE RULE OF PRACTICAL FAIRNESS

Union 208 agreements are extensive because generations of employers and crew members have demonstrated an impressive ability to locate ambiguities.

Yet the guiding standard remains practical fairness.

Ask:

- Was the risk disclosed?
- Was the worker equipped?
- Was the order necessary?
- Was the crew paid?
- Was discipline proportionate?
- Was medical care provided?
- Was rescue attempted?
- Was the contract honored?
- Were changed conditions addressed honestly?
- Did both sides perform their obligations?

Not every dispute requires a tribunal.

Many can be resolved by acknowledging the obvious.

If a captain promised hazard pay for entering an alien ruin and the crew entered the alien ruin, hazard pay is owed.

The fact that no one was consumed by an ancient machine does not retroactively make the mission safe.

XLIII. ADVICE TO CAPTAINS

A captain operating under Union 208 standards should:

- Use written contracts.
- Establish pay clearly.
- Record crew assignments.
- Maintain accurate rosters.
- Listen to safety reports.
- Provide equipment.
- Report changes in mission scope.
- Keep wage records.
- Respect representation.
- Document serious discipline.
- Negotiate before resentment becomes refusal.
- Pay agreed shares.
- Avoid leaving people outside.
- Avoid threatening the union representative beside the reactor.

Union standards do not weaken command.

They clarify it.

A crew that knows the rules is easier to lead than one continually wondering whether the captain intends to cheat, abandon, or execute them.

Trust reduces the amount of force required to maintain order.

It also improves the odds that someone will pull the captain out of a flooded compartment.

XLIV. ADVICE TO CREW MEMBERS

A union crew member should:

- Read the contract.
- Know the wage.
- Understand the mission.
- Perform assigned duties.
- Report shortages.
- Report unsafe conditions.
- Preserve evidence of violations.
- Use the grievance process.
- Support legitimate collective action.
- Avoid turning every personal irritation into a constitutional crisis.
- Remember that the captain may also be a worker trapped under an unreasonable employer.
- Pay union dues where required.
- Assist other mariners in distress.

Solidarity is not the right to demand that everyone else perform your work.

Union protection assumes participation.

A crew member who spends the entire voyage hiding in storage and emerges at port to demand a full salvage share should prepare for a detailed discussion of the attendance records. And the meaning of “equal” compensation, And the consequences from their crew mates .

XLV. THE UNION AND THE DEAD

Union 208 maintains records of lost submariners.

Where possible, it assists in recovering bodies, notifying dependents, transferring wages, and preserving final messages.

Where recovery is impossible, the union may record the location of the loss and mark the route.

Some wreck sites are considered informal graves.

Salvage crews are expected to treat remains with respect.

Necessary equipment may be recovered.

Identification should be preserved.

Personal effects should be returned when practical.

The dead cannot defend their contract rights.

The union does so for them.

This is among the oldest and most solemn of its duties.

XLVI. THE UNION MOTTO

Union 208 has accumulated several mottos.

The formal motto is:

No Mariner Abandoned. No Labor Forgotten.

The dockworkers' version is:

No Pay, No Pumps.

The engineers' version is:

You Can Fire Us After We Keep You Alive.

The captains' version is:

Please Submit Grievances Before Departure.

The Saville memorial uses:

We Heard. We Remember. We Return.

Each contains part of the truth.

XLVII. FINAL DECLARATION

Beneath Europa's ice, survival is collective.

Every pressure door was installed by someone.

Every fuel rod was mined, fabricated, transported, inspected, and loaded by someone.

Every station light depends upon a chain of workers stretching through the darkness.

Every submarine route exists because crews traveled it before charts were reliable and rescue was uncertain.

Standard Union 208 exists because those workers learned that isolated courage was not enough.

A single crew can be threatened.

A single captain can be cheated.

A single vessel can disappear.

A united maritime network cannot be ignored so easily.

The union has survived governments, corporations, cartels, wars, blockades, disasters, schisms, frauds, and generations of administrators confidently predicting that organized labor was no longer necessary.

It remains necessary because Europa remains Europa.

The ice is still thick.

The sea is still dark.

The pressure is still waiting.

The creatures remain hungry.

The contracts remain complicated.

The employers remain creative.

The crews still require food, air, fuel, medicine, pay, rescue, and the knowledge that someone will answer when they call.

Remember the Crampton refusal.

Remember the blacklist years.

Remember the Saville crewmen.

Remember the stations that believed they could function without the people who carried their food and repaired their machinery.

A station may threaten a submarine.

A cartel may threaten a crew.

An employer may threaten a worker.

But when Union 208 withdraws its hands, the cargo remains at the dock, the pumps remain broken, the reactor parts remain undelivered, and the station begins counting how many meals remain.

Do not trifle with the union.

Not because every member is powerful.

Not because every captain agrees.

Not because every station recognizes the same law.

Do not trifle with it because beneath the ice, civilization is a supply chain held together by working people.

When those people refuse to move, very little else does.

THE MARINER'S PLEDGE

I serve beneath the ice by skill, labor, and trust.

I will perform the work I have accepted.

I will preserve the vessel and support its crew.

I will answer distress when I reasonably can.

I will not abandon another mariner for convenience.

I will not accept that desperation removes human dignity.

I will remember those whose sacrifice established these rights.

I will stand with those who keep the routes open, the stations supplied, and the submarines moving.

No mariner abandoned.

No labor forgotten.

We heard.

We remember.

We return.

EXTENDED FOOTNOTE: CONCERNING CULTS, CLOWNS, PARASITES, AND OTHER CONSEQUENCES OF PROLONGED HUMAN ISOLATION

The preceding remarks concerning clown attire should not be interpreted as a complete dismissal of the increasingly serious social phenomenon known as the Children of the Honkmother.

Nor should the good-natured treatment of ordinary shipboard foolishness be mistaken for permission to participate in either of the two major cultic movements currently spreading through European settlements:

- The Children of the Honkmother.
- The Church of the Husk.

The two organizations possess very different beliefs, methods, aesthetics, and preferred forms of catastrophic workplace disruption. They are grouped together here not because they cooperate—they usually do not—but because both represent disturbing responses to the psychological conditions imposed upon humanity beneath Europa's ice.

Every person on Europa lives beneath kilometres of frozen water and crushing pressure.

Every station depends upon machinery that may fail.

Every submarine travels through an environment occupied by creatures that regard humanity as food, competition, contamination, or an unusually noisy snack enclosed in metal.

Everyone knows someone who failed to return.

Everyone has heard a distress signal that ended too suddenly.

Everyone understands that a damaged pump, delayed cargo shipment, depleted fuel reserve, failed harvest, missing medical crate, or cracked pressure seal can convert an ordinary week into a mass-casualty event.

Humanity has not adapted to these conditions with universal dignity.

Some people pray.

Some organize.

Some drink.

Some join the separatists.

Some paint their faces, put horns on their shoes, and declare that the ocean has begun speaking to them through a bicycle horn.

Others willingly infect themselves with a parasite responsible for turning living human beings into hollow, murderous corpses.

The human capacity for coping is broad.

It is not always useful.

THE CHILDREN OF THE HONKMOTHER

The Children of the Honkmother claim to hear a sacred call emanating from the European depths.

This call is variously described as:

- The laughter beneath the ice.
- The Great Honk.
- The Mother's Call.
- The Joyful Pressure.
- The Sound Beyond Sonar.
- A revelation that, when described by less charitable medical officers, bears a striking resemblance to oxygen deprivation.

Members frequently identify themselves through clown masks, greasepaint, colorful clothing, oversized shoes, novelty horns, and an attitude toward restricted machinery best summarized as:

"It would be funny."

The movement began as a mixture of shipboard humor, occupational folklore, countercultural performance, and resistance to the rigid discipline of early Coalition vessels.

In its mildest form, Honkmother tradition consists of jokes, absurd clothing, harmless pranks, mock ceremonies, and efforts to preserve morale aboard vessels where laughter is otherwise in dangerously short supply.

There is nothing inherently criminal about wearing clown clothing.

There is nothing automatically dangerous about carrying a horn.

There is nothing wrong with attempting to make exhausted crew members laugh during a difficult voyage.

The difficulty begins when amusement becomes doctrine.

Radical Honkmother adherents increasingly claim that:

- Ordinary rules are instruments of spiritual confinement.
- Machinery must occasionally be “liberated” through random operation.
- Fear should be cured through surprise.
- Security should be tested to ensure vigilance.
- Dangerous equipment is funniest when used incorrectly.
- Death is merely the final punchline.
- The Honkmother protects those who commit fully to the joke.

Operational records do not support the final claim.

Many submariners regard Honkmother adherents as annoying, disruptive, grease-painted layabouts more interested in jokes, horns, and theatrical nonsense than useful labor.

This reputation is not entirely deserved.

Some adherents are capable assistants, engineers, medics, or security officers who treat clowning as a personal tradition and remain entirely reliable during emergencies.

Others have used “religious observance” to justify:

- Tampering with airlocks.
- Stealing identification cards.
- Mislabeled medical compounds.
- Rewiring alarms.
- Filling diving suits with inconvenient substances.
- Hiding dangerous items in harmless containers.
- Hiding harmless items in dangerous containers.
- Releasing creatures aboard the vessel.
- Disabling essential machinery.
- Administering drugs without consent.
- Escalating practical jokes until the casualty count becomes the punchline.

A joke ceases to be harmless when it interferes with life support, weapons, navigation, medicine, the reactor, hull integrity, or the ability of another crew member to remain alive.

This boundary should not be difficult to understand.

Unfortunately, radical Honkmother theology appears to consider clearly marked boundaries a form of audience participation.

Captains are advised to judge individuals by conduct rather than costume.

A clown who repairs the hull, carries ammunition, obeys recall orders, and performs assigned duties is a crew member with unusual fashion preferences.

A clown who injects the engineer with paralytic agents and opens the ballast hatch because "the timing felt right" is a security matter.

The horn is not the crime.

The flooding is.

THE CHURCH OF THE HUSK

The Church of the Husk presents a substantially darker problem.

Its members revere, accept, or seek union with the parasitic organism commonly called the Husk parasite.

The same organism has killed thousands of submariners directly and contributed to the deaths of many more through abandoned stations, infected wrecks, lost vessels, quarantine failures, and containment operations.

Exact casualty figures remain disputed because several early outbreaks were recorded as station failures, piracy, unexplained crew loss, or "biological irregularities," which is administrative language for discovering that the dead have begun walking.

Husk infection may progressively alter the body, impair communication, compromise ordinary human behavior, and transform the victim into an active threat to surrounding personnel.

The organism is not misunderstood medicine.

It is not a harmless adaptation.

It is not a spiritual vaccination.

It is not an unusually committed dietary preference.

It is a lethal parasite.

Church adherents nevertheless describe infection as:

- Communion with Europa.
- Freedom from fear.
- Adaptation to the sea.
- The next stage of humanity.
- Release from the weakness of ordinary flesh.
- Proof that Europa has chosen them.

Medical personnel generally describe it as:

- Infection.
- Parasitic compromise.
- A quarantine emergency.
- A reason to lock the medical cabinet.
- A reason to ask why the assistant has several unlabeled syringes.

The Church is more secretive than the Honkmother movement.

Its members may conceal their affiliation, hide biological material, sabotage medical examinations, interfere with treatment, or deliberately expose other crew members without consent.

Some cells believe forced infection is an act of liberation.

Others view uninfected humanity as temporary, inferior, or spiritually deaf.

Still others appear less interested in religion than in using the Church as cover for assassination, sabotage, biological warfare, or personal experimentation.

Any crew member suspected of deliberate Husk exposure must be treated as both a medical and security concern.

This does not permit immediate execution of anyone who coughs.

It does require:

- Prompt medical evaluation.
- Isolation when justified.
- Control of biological samples.
- Inspection of medical storage.
- Protection of treatment supplies.
- Reporting unexplained symptoms.
- Preventing infected personnel from accessing crowded areas.
- Avoiding the traditional mistake of waiting until the patient has fully transformed because confronting them earlier felt impolite.

An infected person may still be a patient.

A cultist deliberately spreading infection is a hostile actor.

Medical and security personnel must understand the distinction, and must also understand the moment at which the first condition becomes the second.

A willingness to expose oneself to the Husk does not create the right to expose everyone else.

Consent does not spread through ventilation.

WHY THESE MOVEMENTS EXIST

It is convenient to dismiss both groups as collections of fools and madmen.

There are certainly fools and madmen among them.

There are also frightened laborers, bereaved families, abandoned colonists, traumatized survivors, disillusioned station personnel, and people searching for meaning inside a civilization whose foundations have become alarmingly uncertain.

Europa was not originally built as a self-sufficient human world.

The colonies developed as extensions of Earth's economy.

Stations depended upon regular shipments of:

- Precision machinery.
- Pharmaceuticals.
- Electronic components.
- Industrial chemicals.
- Finished tools.
- Agricultural nutrients.
- Replacement membranes.
- Reactor components.
- Processed food.
- Genetic material.
- Specialized personnel.
- Thousands of ordinary goods that become noticeable only after the last shipment fails to arrive.

European settlement was built upon the assumption that Earth would remain at the other end of the supply line.

When that supply relationship became unreliable, delayed, politically restricted, or functionally severed, European society discovered that it possessed a collection of remote commercial outposts rather than a complete civilization.

A market can distribute finished goods.

It cannot distribute goods that no one is capable of manufacturing.

Credits do not become bearings.

Contracts do not become antibiotics.

Stock valuations do not become oxygen membranes.

A station may contain several wealthy investors and still die because nobody possesses the tooling required to fabricate a replacement pump shaft.

The Coalition has therefore been forced to undertake a monumental reconstruction of European industry.

It must create, expand, or reorganize:

- Foundries.
- Machine works.
- Chemical plants.
- Agricultural systems.
- Pharmaceutical production.
- Material extraction.
- Refining.
- Toolmaking.
- Component standardization.
- Transport networks.
- Training systems.
- Labor allocation.
- Repair depots.
- Emergency reserves.
- Every other piece of industrial civilization Earth previously supplied from a distance.

This is not merely economic adjustment.

It is the attempted creation of a self-sustaining industrial ecology beneath an alien ocean.

Coalition administrators have described the process as rational consolidation, emergency coordination, strategic production, and temporary allocation.

Workers have more commonly described it as:

“We still do not have enough food.”

The early European economy was structured around commercial extraction, scientific prestige, private ownership, competitive contracts, and continuing imports from Earth.

When those imports failed, the same institutions were suddenly expected to centrally coordinate production across a society they had never designed to support independently.

Many administrators understood finance.

Some understood logistics.

Far fewer understood how to build an economy in which raw ore enters one end and a functioning pressure valve emerges from the other without relying upon six companies and three planets that are no longer answering messages.

The resulting shortages have been severe.

Whole stations have repeatedly approached famine.

Medical systems have run out of basic compounds.

Reactors have operated on improvised fuel.

Pumps have remained in service long after replacement should have been mandatory.

Workers have been ordered to accept lower rations while administrators promised that the next convoy would solve everything.

The next convoy has developed a reputation for not arriving.

THE JOVIAN SEPARATISTS

The rise of the Jovian Separatists cannot be understood separately from these shortages.

They are not a cult in the same sense as the Children of the Honkmother or the Church of the Husk.

They are a revolutionary political movement composed largely of workers, miners, technicians, transport crews, station residents, local militias, and people who believe the Coalition has failed to provide survivable conditions.

The movement contains idealists, labor organizers, local autonomists, militants, opportunists, smugglers, extremists, and individuals whose political theory consists primarily of knowing that their children are hungry.

Many separatists possess only a limited understanding of planetary-scale industrial planning.

This is also true of many Coalition officials.

The difference is that separatist workers rarely claim to possess warehouses that do not exist.

Their grievances commonly include:

- Chronic food shortages.
- Unequal rationing.
- Unsafe workplaces.
- Unpaid or devalued wages.
- Preferential supply to wealthy stations.
- Corporate control of essential infrastructure.
- Failure to replace damaged machinery.
- Forced labor through debt.
- Abandonment of frontier settlements.
- Coalition control without reliable Coalition support.

A mechanic does not need a complete theory of macroeconomics to understand that the station requires replacement seals.

A miner does not need to model interstation trade to understand that their family is receiving fewer rations while export cargo continues leaving the dock.

A submarine crew does not need an advanced education in industrial policy to notice that their vessel has been sent through hostile water without sufficient fuel.

The Separatists have turned these conditions into a political movement.

Whether they can create a more functional society remains unresolved.

Knowing that the present system is failing does not automatically provide the expertise required to replace it.

Revolutions are exceptionally good at identifying who should no longer be in charge.

They are less consistent at manufacturing pump components.

THE SUBMARINER BETWEEN FACTIONS

Ordinary submariners often find themselves operating between Coalition stations, separatist territories, corporate enclaves, cartel ports, union docks, Honkmother gatherings, and communities where the local authorities insist no cults are present despite the chanting audible through the walls.

The general crewman should therefore understand several principles.

First, clothing alone does not establish guilt.

Second, concealed parasites are more concerning than visible face paint.

Third, visible face paint does not make sabotage acceptable.

Fourth, political affiliation does not automatically make someone an enemy combatant.

Fifth, anyone attempting to disable life support, spread infection, seize the vessel, or kill the crew has moved beyond philosophical disagreement.

Captains and security officers should avoid treating every clown as a cultist, every dissident as a separatist agent, every religious person as a Husk adherent, or every shortage complaint as the beginning of mutiny.

Broad suspicion creates the exact isolation and resentment from which radical movements recruit.

Equally, the crew should not dismiss obvious warning signs merely because confrontation may be uncomfortable.

Warning signs include:

- Unexplained biological samples.
- Hidden syringes.
- Missing medical supplies.
- Deliberate infection.

- Repeated tampering with life support.
- Cult markings placed inside restricted compartments.
- Attempts to isolate vulnerable crew members.
- Sabotage described as ritual.
- Violence described as comedy.
- Claims that the submarine must be sacrificed.
- Claims that the crew must “hear the call” through depressurization.

The ocean already possesses enough mechanisms for killing submariners.

It does not require assistance from theology.

A NOTE TO ASSISTANTS

Assistants are especially vulnerable to recruitment by these groups.

The reason is not that assistants are foolish.

It is that inexperienced general crewmen may feel overlooked, underqualified, or uncertain of their place aboard the submarine.

Both movements offer immediate identity.

The Honkmother offers belonging through humor, performance, irreverence, and the promise that being excluded from formal authority makes one spiritually superior to it.

The Church of the Husk offers belonging through transformation, secrecy, certainty, and the promise that human vulnerability can be escaped rather than endured.

The healthier answer is less dramatic:

Learn the vessel.

Support the crew.

Develop useful skills.

Find people who value your work.

Join a union.

Wear a funny hat without turning it into a theology.

You do not require a parasite to become important.

You do not require sabotage to become memorable.

You do not need to hear a supernatural horn from the abyss to justify having a personality.

A competent assistant who carries ammunition, repairs pumps, recovers casualties, supports away teams, and keeps everyone laughing at appropriate times contributes more to human survival than a hundred cultists making declarations beside an open airlock.

This may not sound as mystical.

It has the advantage of working.

FINAL CAUTION

The Children of the Honkmother and the Church of the Husk are symptoms of a civilization under extreme pressure.

One turns despair into absurdity.

The other turns despair into surrender.

Both can become lethal when private belief is used to justify harm to the crew.

Europa does not care whether the person destroying a submarine is doing so for money, politics, religion, comedy, spiritual transformation, or because a voice in the ocean promised applause.

The hull ruptures in precisely the same way.

The captain is therefore advised to permit harmless eccentricity, protect legitimate belief, listen to political grievances, and act firmly against sabotage, coercion, deliberate infection, or violence.

Let the harmless clown honk.

Let the worker complain.

Let the separatist explain why the ration system is failing.

Let the religious crew member pray.

But when someone reaches for the reactor controls, the Husk syringe, or the emergency ballast valve while announcing that "the time has come," security may reasonably conclude that the discussion has entered its practical phase.

Humanity beneath the ice has enough enemies.

It should exercise care before inventing additional ones.